Course Notes

Abstract Algebra III

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March 22, 2026

Contents

Chapter 1

Lecture 1: Modules

Message: QQ, BB. HW 30%, Mid 20%, Final 50%

1.1 Basic Definitions and Examples

Definition 1.1.1. *Let R be a ring with 1, and let M be an abelian group. We say that M is a left R-module if R "left acts" on M:*

$$R \times M \to M, \quad (r, m) \mapsto rm$$

such that

$$\begin{aligned} r(m+n) &= rm + rn, \\ (r+s)m &= rm + sm, \\ (rs)m &= r(sm), \\ 1m &= m. \end{aligned} \quad \text{for any } r, s \in R, m, n \in M$$

*A **right** R-**module** is defined by $(r, m) \mapsto mr$ with the axiom $m(rs) = (mr)s$. If R is commutative, left-modules and right-modules coincide. Otherwise, one can transform using the opposite ring R^{op}.*

Consider M an R-module, then $\forall r \in R$, $\varphi_r : M \to M$ by $\varphi_r(m) = rm$ is an abelian group homomorphism. Let

$$\mathrm{End}(M) = \{\varphi : M \to M \text{ is a abelian group homomorphism } \}$$

Then $\mathrm{End}(M)$ is a ring with composition as the multiplication.

Theorem 1.1.2. *Let M be an abelian group. Then M is an R-module if and only if there is a ring homomorphism $R \to \mathrm{End}(M)$.*

Proof. $(\Rightarrow)$ Suppose M is an R-module. Define $\rho : R \to \mathrm{End}(M)$ by

$$\rho(r)(m) = rm, \quad \forall r \in R, m \in M.$$

For each $r \in R$, the map $\rho(r) : M \to M$ is a group homomorphism since $r(m_1 + m_2) = rm_1 + rm_2$. Moreover, ρ is a ring homomorphism:

- $\rho(r+s)(m) = (r+s)m = rm + sm = \rho(r)(m) + \rho(s)(m)$, so $\rho(r+s) = \rho(r) + \rho(s)$.
- $\rho(rs)(m) = (rs)m = r(sm) = \rho(r)(\rho(s)(m))$, so $\rho(rs) = \rho(r) \circ \rho(s)$.
- $\rho(1)(m) = 1m = m$, so $\rho(1) = \mathrm{id}_M$.

($\Leftarrow$) Suppose $\rho : R \to \mathrm{End}(M)$ is a ring homomorphism. Define the scalar multiplication $R \times M \to M$ by

$$(r, m) \mapsto \rho(r)(m).$$

We verify the module axioms:

- $r(m_1 + m_2) = \rho(r)(m_1 + m_2) = \rho(r)(m_1) + \rho(r)(m_2) = rm_1 + rm_2$.
- $(r+s)m = \rho(r+s)(m) = (\rho(r) + \rho(s))(m) = \rho(r)(m) + \rho(s)(m) = rm + sm$.
- $(rs)m = \rho(rs)(m) = (\rho(r) \circ \rho(s))(m) = \rho(r)(\rho(s)(m)) = r(sm)$.
- $1m = \rho(1)(m) = \mathrm{id}_M(m) = m$.

Hence M is an R-module. □

Example 1.1.3. *(1) A vector space is a module over a field.*
(2) An abelian group is a module over the ring of integers $\mathbb{Z}$.

$$\mathbb{Z} \times M \to M, \quad (n, m) \mapsto n \cdot m := m + m + \cdots + m \text{ (n times)}$$

(3) An ideal I of a ring R is a module over R.
(4) Let M be an abelian group. Then M is an $\mathrm{End}(M)$-module via

$$\mathrm{End}(M) \times M \to M, \quad (\varphi, m) \mapsto \varphi(m).$$

(5) Let V be a vector space over a field K, and let $A : V \to V$ be a linear transformation. Then V is a $K[x]$-module via

$$K[x] \times V \to V, \quad (f(x), v) \mapsto f(A)v.$$

(6) If $|M| = 1$, then M is called the zero module.
(7) Let R be a ring with 1 and $A \in M_n(R)$. The solution of $AX = 0$ where $X \in R^n$ is a right R-module.

Definition 1.1.4 (R-submodule)**.** *Let R be a ring (with 1) and let M be a left R-module. A subset $N \subseteq M$ is called an R-submodule of M if:*

1. $0 \in N$;
2. $n_1, n_2 \in N \implies n_1 + n_2 \in N$;

3. $n \in N \implies -n \in N$;

4. $r \in R,\ n \in N \implies rn \in N$.

Equivalently, N is a subgroup of the abelian group $(M, +)$ that is closed under the R-action.

Example 1.1.5. *(1) Let V be a vector space over a field K, then any subspace of V is a k-submodule of V.*
(2) Let G be an abelian group, then any subgroup of G is a $\mathbb{Z}$-submodule of G.
(3) Let R be a ring, ${}_RR$ is a left (regular) R-module. Then any left ideal of R is a R-submodule of ${}_RR$.
(4) Let G be an abelian group, which is an $\mathrm{End}(G)$-module. Then any full invariant subgroup of G is a $\mathrm{End}(G)$-submodule of G.

Question: Are submodules of a finitely generated R-module finitely generated?
Answer: No, not in general. This property holds if and only if the ring R is Noetherian.

Example 1.1.6 (Submodule of a f.g. module that is not f.g.)**.** *Let k be a field and consider the polynomial ring $R = k[x_1, x_2, x_3, \dots]$ in infinitely many variables.*

- *The ring R itself can be viewed as a left R-module, which we denote by $M = R$. M is finitely generated, for instance by the single element $1 \in R$.*
- *Consider the ideal $I = \langle x_1, x_2, x_3, \dots \rangle$ generated by all the variables. This ideal I is a submodule of $M = R$.*
- *This submodule I is not finitely generated. To see this, suppose for contradiction that I is finitely generated by a set of polynomials $\{f_1, \dots, f_n\}$. Let N be the largest index of any variable appearing in any of the polynomials $f_1, \dots, f_n$. Then any element in the ideal generated by $\{f_1, \dots, f_n\}$ must be a polynomial in variables $x_1, \dots, x_N$. However, the polynomial x_{N+1} is in I but cannot be expressed as a combination of $f_1, \dots, f_n$. This is a contradiction.*

Therefore, the ideal I is a submodule of the finitely generated R-module R which is not itself finitely generated. This shows that the ring R is not Noetherian.

Definition 1.1.7. *Let M be an R-module, $M_i, i \in I$ are R-submodules of M. Then*

$$\bigcap_{i \in I} M_i$$

is an R-submodule of M.

$$\sum_{i \in I} = \{ \sum_{i \in I \ finite} m_i | m_i \in M_i \}$$

is an R-submodule of M.

Definition 1.1.8. *Let M be an R-module, $S \subseteq M$ a subset.*

$$< S >= \{ \sum_{finite} a_i s_i | a_i \in R, s_i \in S \}$$

or

$$< S >= \bigcap_{N is\ an\ R-submodule\ of\ M,\ S \subseteq N} N$$

is an R-submodule of M, called the R-submodule of M generated by S.
If $M =< S >$, then we say M is generated by S. In particular, if $|S| = 1$ then we call M a cyclic module. If $|S| < \infty$ then we call M a finite generated module.

Definition 1.1.9. *Let R be a ring and let M be a left R-module. We say that M is a* Noetherian module *if every ascending chain of submodules*

$$M_1 \subseteq M_2 \subseteq M_3 \subseteq \cdots$$

stabilizes; that is, there exists $n \geq 1$ such that

$$M_n = M_{n+1} = M_{n+2} = \cdots .$$

Equivalently, M is Noetherian if every submodule of M is finitely generated.

1.2 Quotient Modules and Homomorphism Theorems

Definition 1.2.1. *Let R be a ring, let M be a left R-module, and let N be a submodule of M. The* quotient module *(or* factor module*) of M by N is the quotient abelian group*

$$M/N = \{ x + N \mid x \in M \},$$

with scalar multiplication defined by

$$r(x + N) = rx + N \qquad (r \in R,\ x \in M).$$

This is well-defined, and with this operation M/N becomes a left R-module.

Example 1.2.2. *(1) Let R be a ring and an R-module, I be an ideal of R, then R/I is a quotient R-module of R by I.*
(2) Consider the ring $\mathbb{Z}$ and the module $\mathbb{Z}$ over $\mathbb{Z}$, then for $n \in \mathbb{Z}$, $n\mathbb{Z}$ is a submodule of $\mathbb{Z}$, and $\mathbb{Z}/n\mathbb{Z}$ is a quotient $\mathbb{Z}$-module of $\mathbb{Z}$ by $n\mathbb{Z}$. Furthermore, $\mathbb{Z}/n\mathbb{Z}$ is a cyclic module.

Definition 1.2.3. *Let R be a ring, and let M and N be left R-modules. A group homomorphism*

$$f : M \to N$$

*is called an R-*module homomorphism *(or R-*linear map*) if for all $x, y \in M$ and all $r \in R$, one has*

$$f(rx) = rf(x) \quad \forall r \in R, x \in M.$$

Definition 1.2.4. *Let f be an R-module homomorphism. The* kernel *of f is the set*

$$\ker(f) = \{x \in M \mid f(x) = 0\}.$$

$\ker(f)$ *is a submodule of M. The* image *of f is the set*

$$\operatorname{im}(f) = \{f(x) \in N \mid x \in M\}.$$

$\operatorname{im}(f)$ *is a submodule of N.*
If $\ker(f) = 0$*, then f is injective. If* $\operatorname{im}(f) = N$*, then f is surjective. If* $\ker(f) = 0$ *and* $\operatorname{im}(f) = N$*, then f is bijective.*
A bijective R-module homomorphism is called an isomorphism.

Example 1.2.5. *Let R be a ring and M, N be an R-modules. Then* $\operatorname{Hom}_R(M, N)$ *is an R-module when R is commutative.*

Theorem 1.2.6. *If $f : M \to N$ is an R-module homomorphism, then we have a canonical R-module isomorphism:*

$$M/\ker(f) \to \operatorname{im}(f)$$

Theorem 1.2.7. *Let N be an R-submodule of M, $\pi_N : M \to M/N$ be a quotient map. Then π_N induces a natural bijection:*

$$\{R-\textit{submodules of } M \textit{ containing } N\} \to \{R-\textit{submodules of } M/N\}$$

Theorem 1.2.8. *If $N \subseteq H$ are two R-submodules of M, then*

$$M/H \cong M/N/(H/N),$$

Theorem 1.2.9. *If N, H are R-submodules of M, then*

$$\frac{H+N}{N} \cong \frac{H}{H \cap N}$$

Proposition 1.2.10. *Let M be an R-module, N a submodule of M, M' an R-module, and $\varphi : M \to M'$ a morphism such that $N \subseteq \ker(\varphi)$. Then there exists a unique morphism*

$$\varphi_N : M/N \to M'$$

such that the following diagram commutes. Moreover, $\ker(\varphi_N) = \ker \varphi / N$.

$$\begin{tikzcd} M \arrow[r, "\varphi"] \arrow[d, "\pi_N"'] & M' \\ M/N \arrow[ur, dashed, "\varphi_N"'] & \end{tikzcd}$$

Remark 1.2.11. *The pair $(M/N, \pi_N)$ is unique in the following sense: if (S, θ) is a pair such that*

- *S is an R-module,*
- *$\theta : M \to S$ is a surjective morphism satisfying the property of $(M/N, \pi_N)$ described in the above proposition,*

then there exists a unique isomorphism $\psi : M/N \to S$ such that the following diagram commutes.

$$\begin{tikzcd} M \arrow[r, two heads, "\theta"] \arrow[d, two heads, "\pi_N"'] & S \\ M/N \arrow[ur, dashed, "\exists!\,\psi"'] & \end{tikzcd}$$

Proof. Since (S, θ) satisfies the same universal property as $(M/N, \pi_N)$, we may apply that property to the canonical projection

$$\pi_N : M \to M/N.$$

Because $N \subseteq \ker(\pi_N)$, there exists a unique morphism

$$u : S \to M/N$$

such that

$$u \circ \theta = \pi_N.$$

On the other hand, since $(M/N, \pi_N)$ has the universal property of the quotient by N, and since $N \subseteq \ker(\theta)$, there exists a unique morphism

$$\psi : M/N \to S$$

such that

$$\psi \circ \pi_N = \theta.$$

We now show that ψ and u are inverse to each other.

First, compute

$$(u \circ \psi) \circ \pi_N = u \circ (\psi \circ \pi_N) = u \circ \theta = \pi_N.$$

But the identity morphism $\mathrm{id}_{M/N} : M/N \to M/N$ also satisfies

$$\mathrm{id}_{M/N} \circ \pi_N = \pi_N.$$

By the uniqueness part of the universal property of $(M/N, \pi_N)$, it follows that

$$u \circ \psi = \mathrm{id}_{M/N}.$$

Similarly,

$$(\psi \circ u) \circ \theta = \psi \circ (u \circ \theta) = \psi \circ \pi_N = \theta.$$

But the identity morphism $\mathrm{id}_S : S \to S$ also satisfies

$$\mathrm{id}_S \circ \theta = \theta.$$

By the uniqueness part of the universal property of (S, θ), we obtain

$$\psi \circ u = \mathrm{id}_S.$$

Therefore ψ is an isomorphism, with inverse u.

Finally, the uniqueness of ψ follows directly from the universal property of $(M/N, \pi_N)$: indeed, ψ is the unique morphism

$$\psi : M/N \to S$$

such that

$$\psi \circ \pi_N = \theta.$$

Hence there exists a unique isomorphism

$$\psi : M/N \to S$$

for which the diagram commutes. □

Proposition 1.2.12. *There is a unique isomorphism $\bar{\varphi} : \mathrm{coim}\,\varphi \to \mathrm{im}\,\varphi$ such that the following diagram commutes:*

$$\begin{tikzcd}
M \arrow[r, "\varphi"] \arrow[d, "\pi_{\ker(\varphi)}"'] & M' \\
\mathrm{coim}(\varphi) \arrow[r, "\overline{\varphi}"] & \mathrm{im}(\varphi) \arrow[u, hook]
\end{tikzcd}$$

Recall:

$$\begin{aligned} &\varphi : M \to M', \quad \textit{R-morphism} \\ &\operatorname{coker}(\varphi) = M'/\operatorname{im}(\varphi) \\ &\operatorname{coim}(\varphi) = M/\ker\varphi \end{aligned}$$

Definition 1.2.13. *Let M be an R-module and I an ideal of R. Denote by IM the submodule of M generated by all elements am where $a \in I$, $m \in M$.*

$$IM = \left\{ \sum_{\textit{finite}} a_i m_i \mid a_i \in I,\ m_i \in M \right\}$$

Proposition 1.2.14. *Let M be an R-module. Then $R \to \operatorname{End}(M)$ is a ring homomorphism.*

1. *$R \to \operatorname{End}(M)$ factors through the natural surjection $R \to R/I$ if and only if $IM = 0$:*

$$\begin{tikzcd} R \arrow[rr, "\rho"] \arrow[dr, two heads, "\pi_I"'] & & \operatorname{End}(M) \\ & R/I \arrow[ur, dashed, "\exists\,\rho_I"'] & \end{tikzcd}$$

   Where ρ factors through π_I means $\exists \rho_I : R/I \to \operatorname{End}(M)$ such that the diagram above commutes.

2. *The module M/IM is naturally endowed with a structure of R/I-module.*

3. *Let $\varphi : M \to M'$ be an R-homomorphism. Then $\varphi(IM) \subseteq IM'$, so φ induces a morphism of R/I-modules*

$$M/IM \to M'/IM'.$$

4. *If $M = M_1 \oplus \cdots \oplus M_n$, then $M/IM = M_1/IM_1 \oplus \cdots \oplus M_n/IM_n$.*

5. *$R^n/IR^n \cong (R/IR)^n$.*

Proof. **(1)** Suppose first that ρ factors through π_I. Then there exists a ring homomorphism

$$\rho_I : R/I \to \operatorname{End}_{\mathbb{Z}}(M)$$

such that

$$\rho = \rho_I \circ \pi_I.$$

If $a \in I$, then $\pi_I(a) = 0$, so

$$\rho(a) = \rho_I(0) = 0.$$

Therefore $am = 0$ for every $m \in M$. Since IM is generated by elements of the form am with $a \in I$ and $m \in M$, it follows that

$$IM = 0.$$

Conversely, assume that $IM = 0$. Then for every $a \in I$ and every $m \in M$ we have

$$\rho(a)(m) = am = 0.$$

Thus $\rho(a) = 0$, so $I \subseteq \ker \rho$. Hence ρ factors uniquely through the quotient ring R/I: there exists a unique ring homomorphism

$$\rho_I : R/I \to \mathrm{End}_{\mathbb{Z}}(M)$$

such that

$$\rho = \rho_I \circ \pi_I.$$

(2) Define an action of R/I on M/IM by

$$(r + I) \cdot (m + IM) := rm + IM.$$

We must show that this is well-defined.

If $r + I = r' + I$, then $r - r' \in I$, so

$$(r - r')m \in IM,$$

hence

$$rm + IM = r'm + IM.$$

If $m + IM = m' + IM$, then $m - m' \in IM$. Since IM is a submodule of M, we have

$$r(m - m') \in IM,$$

so

$$rm + IM = rm' + IM.$$

Thus the action is well-defined. The module axioms follow immediately from the module axioms on M. Therefore M/IM is naturally an R/I-module.

(3) Let $\varphi : M \to M'$ be an R-module homomorphism. Take any element of IM, say

$$x = \sum_{j=1}^{t} a_j m_j \qquad (a_j \in I,\ m_j \in M).$$

Then

$$\varphi(x) = \sum_{j=1}^{t} \varphi(a_j m_j) = \sum_{j=1}^{t} a_j \varphi(m_j).$$

Since each $a_j \in I$ and each $\varphi(m_j) \in M'$, it follows that

$$a_j \varphi(m_j) \in IM'.$$

Hence $\varphi(x) \in IM'$, and therefore

$$\varphi(IM) \subseteq IM'.$$

Thus we may define

$$\overline{\varphi} : M/IM \to M'/IM', \qquad \overline{\varphi}(m + IM) = \varphi(m) + IM'.$$

This is well-defined: if $m + IM = m' + IM$, then $m - m' \in IM$, so

$$\varphi(m) - \varphi(m') = \varphi(m - m') \in IM',$$

hence

$$\varphi(m) + IM' = \varphi(m') + IM'.$$

Finally, $\overline{\varphi}$ is R/I-linear, since

$$\overline{\varphi}\big((r+I)\cdot(m+IM)\big) = \overline{\varphi}(rm+IM) = \varphi(rm)+IM' = r\varphi(m)+IM' = (r+I)\cdot\overline{\varphi}(m+IM).$$

(4) Assume that

$$M = M_1 \oplus \cdots \oplus M_n.$$

We first show that

$$IM = IM_1 \oplus \cdots \oplus IM_n.$$

If $x \in IM$, then x is a finite sum of elements of the form am with $a \in I$ and $m \in M$. Writing

$$m = m_1 + \cdots + m_n \qquad (m_i \in M_i),$$

we get

$$am = am_1 + \cdots + am_n,$$

and each $am_i \in IM_i$. Hence $x \in IM_1 + \cdots + IM_n$, so

$$IM \subseteq IM_1 + \cdots + IM_n.$$

The reverse inclusion is immediate because each $M_i \subseteq M$, hence each $IM_i \subseteq IM$. Therefore

$$IM = IM_1 + \cdots + IM_n.$$

To see that the sum is direct, suppose

$$x_1 + \cdots + x_n = 0 \qquad \text{with } x_i \in IM_i \subseteq M_i.$$

Since

$$M = M_1 \oplus \cdots \oplus M_n$$

is a direct sum, we must have $x_i = 0$ for all i. Thus

$$IM = IM_1 \oplus \cdots \oplus IM_n.$$

Now define

$$\Psi : M_1/IM_1 \oplus \cdots \oplus M_n/IM_n \to M/IM$$

by

$$\Psi(m_1 + IM_1, \ldots, m_n + IM_n) = (m_1 + \cdots + m_n) + IM.$$

This map is well-defined and R/I-linear. It is surjective because every element of M can be written as $m_1 + \cdots + m_n$. It is injective because

$$(m_1 + \cdots + m_n) + IM = IM$$

means that

$$m_1 + \cdots + m_n \in IM = IM_1 \oplus \cdots \oplus IM_n,$$

so each $m_i \in IM_i$. Hence

$$m_i + IM_i = IM_i$$

for all i, and the kernel is trivial. Therefore Ψ is an isomorphism:

$$M/IM \cong M_1/IM_1 \oplus \cdots \oplus M_n/IM_n.$$

(5) Since

$$R^n = R \oplus \cdots \oplus R$$

(n copies), part (4) yields

$$R^n/IR^n \cong R/IR \oplus \cdots \oplus R/IR.$$

Therefore

$$R^n/IR^n \cong (R/IR)^n.$$

This completes the proof. □

1.3 Direct Sums and Free Modules

Definition 1.3.1. *Direct sum and direct product* $\left(\bigoplus_{i\in I} M_i \text{ \& } \prod_{i\in I} M_i\right)$.
Let $(M_i)_{i\in I}$ *be a family of* R*-modules. The direct product of* $(M_i)_{i\in I}$ *is the module*

$$\prod_{i\in I} M_i = \{\varphi : I \to \cup_{i\in I} M_i | \forall i \in I, \varphi(i) \in M_i\}$$

with the componentwise addition and scalar multiplication. i.e.

$$R \times \prod_{i\in I} M_i \to \prod_{i\in I} M_i, \quad (r,\varphi) \mapsto (r\varphi(i))_{i\in I}$$

The direct sum of $(M_i)_{i\in I}$ *is the module*

$$\bigoplus_{i\in I} M_i = \{\varphi \in \prod_{i\in I} M_i | \sharp\{i \in I|\varphi(i) \neq 0\} < \infty\}$$

Equal when $|I| < \infty$*. In a module category, direct product = product, direct sum = coproduct.*

Theorem 1.3.2. *Suppose that* $M_1, \dots, M_n$ *are* R*-submodules of* M *in which* $M = M_1 + \cdots + M_n$, $I = [n]$*. Then the followings are equivalent:*

1. *The map* $\varphi : \sum_{i\in I} M_i \to M$ *by* $\varphi((m_i)_{i\in I}) = \sum_{i\in I} m_i, m_i \in M_i$ *is an isomorphism.*
2. *Any element of* M *can be uniquely written as a sum of elements of* M_i*.*
3. 0 *can be uniquely written as a sum of elements of* M_i*. More precisely,* $0 = 0 + \ldots + 0$*.*
4. *For* $i \in I$, $M_i \cap (\sum_{j\neq i} M_j) = 0$*.*

Proof. We prove

$$(1) \Rightarrow (2) \Rightarrow (3) \Rightarrow (4) \Rightarrow (1).$$

$(1) \Rightarrow (2)$. If φ is an isomorphism, then for every $m \in M$ there exists a unique

$$(m_1, \dots, m_n) \in \sum_{i=1}^{n} M_i$$

such that

$$m = \varphi(m_1, \dots, m_n) = m_1 + \cdots + m_n.$$

Hence every element of M can be written uniquely as a sum of elements of the M_i.

$(2) \Rightarrow (3)$. Apply (2) to the element $0 \in M$.

$(3) \Rightarrow (4)$. Fix $i \in I$, and let

$$x \in M_i \cap \sum_{j\neq i} M_j.$$

Then $x \in M_i$, and there exist $m_j \in M_j$ for $j \neq i$ such that

$$x = \sum_{j\neq i} m_j.$$

Therefore

$$0 = x - \sum_{j \neq i} m_j$$

is a decomposition of 0 as a sum of elements from the submodules $M_1, \ldots, M_n$. By the uniqueness in (3), this decomposition must be the trivial one. In particular,

$$x = 0.$$

Hence

$$M_i \cap \sum_{j \neq i} M_j = 0.$$

(4) $\Rightarrow$ (1). Since

$$M = M_1 + \cdots + M_n,$$

the map φ is surjective. It remains to show that φ is injective.

Suppose that

$$\varphi(m_1, \ldots, m_n) = 0,$$

that is,

$$m_1 + \cdots + m_n = 0 \qquad \text{with } m_i \in M_i.$$

For each $i \in I$, we have

$$m_i = -\sum_{j \neq i} m_j \in \sum_{j \neq i} M_j.$$

Since also $m_i \in M_i$, it follows that

$$m_i \in M_i \cap \sum_{j \neq i} M_j = 0.$$

Thus $m_i = 0$ for all i, so

$$\ker \varphi = 0.$$

Therefore φ is injective, and hence an isomorphism.

This proves that the four conditions are equivalent. $\square$

Definition 1.3.3 (free module)**.** *Let M be an R-module.*

1. *$x_1, \ldots, x_n \in M$ are linearly independent if*

$$a_1 x_1 + a_2 x_2 + \cdots + a_n x_n = 0 \Rightarrow a_1 = a_2 = \cdots = a_n = 0.$$

2. *M is called free if it is generated by a linearly independent subset. This subset is called a basis of M.*

3. *If M is an R-module with a basis $x_1, \ldots, x_n$, then $M = \langle x_1 \rangle \oplus \cdots \oplus \langle x_n \rangle$.*

Example 1.3.4. $R \oplus \cdots \oplus R = R^n$.

Theorem 1.3.5. *Let R be a commutative ring with 1. Let M be a finitely generated free R-module. Then any basis of M has the same cardinality. The number of elements in a basis is called the rank of the free module.*

Proof. Let $B = \{x_i\}_{i \in I}$ be a basis of M. We first show that any basis is finite. Since M is finitely generated, choose generators $y_1, \ldots, y_n$ of M. Each y_k is a finite R–linear combination of elements of B, so there exists a finite subset $I_0 \subset I$ such that

$$y_1, \ldots, y_n \in \sum_{i \in I_0} Rx_i.$$

Hence

$$M = \sum_{k=1}^{n} Ry_k \subseteq \sum_{i \in I_0} Rx_i \subseteq M,$$

so $M = \sum_{i \in I_0} Rx_i$. If $I \setminus I_0 \neq \varnothing$, pick $j \in I \setminus I_0$. Then

$$x_j \in M = \sum_{i \in I_0} Rx_i,$$

contradicting that B is linearly independent. Thus $I = I_0$ is finite.

Now let $x_1, \ldots, x_r$ and $y_1, \ldots, y_s$ be two bases of M. Let $J \subset R$ be a maximal ideal and set $F := R/J$, a field. Then M/JM is naturally an F-vector space. Write $\overline{m}$ for the class of $m \in M$ in M/JM, and $\overline{a}$ for the class of $a \in R$ in F. We claim that $\overline{x_1}, \ldots, \overline{x_r}$ form an F–basis of M/JM.

Spanning. For any $m \in M$, write $m = \sum_{i=1}^{r} a_i x_i$. Then

$$\overline{m} = \sum_{i=1}^{r} \overline{a_i}\, \overline{x_i},$$

so $\overline{x_1}, \ldots, \overline{x_r}$ span M/JM.

Linear independence. Suppose

$$\sum_{i=1}^{r} \overline{a_i}\, \overline{x_i} = 0 \quad \text{in } M/JM.$$

Then $\sum_{i=1}^{r} a_i x_i \in JM$, so there exist $a_1', \ldots, a_r' \in J$ such that

$$\sum_{i=1}^{r} a_i x_i = \sum_{i=1}^{r} a_i' x_i.$$

Hence

$$\sum_{i=1}^{r} (a_i - a_i')\, x_i = 0.$$

By linear independence of $x_1, \ldots, x_r$, we get $a_i - a_i' = 0$ for all i, so $a_i \in J$, thus $\overline{a_i} = 0$ for all i. Therefore $\overline{x_1}, \ldots, \overline{x_r}$ are F-linearly independent, and form an F–basis of M/JM.

By the same argument, $\overline{y_1}, \ldots, \overline{y_s}$ also form an F–basis of M/JM. Hence

$$r = \dim_F(M/JM) = s.$$

Therefore any two bases of M have the same cardinality. $\square$

Chapter 2

Lecture 2: Tensor Product

2.1 Annihilator, Torsion and Projectivity of Free Modules

Definition 2.1.1. *Let M be an R-module, $m \in M$, $Rm =< m >= \{rm \mid r \in R\}$. Define the annihilator of m in R as:*

$$\mathrm{Ann}_R(m) = \{r \in R \mid rm = 0\}.$$

It is a left ideal of R.

Then we have $R/\mathrm{Ann}_R(m) \cong Rm$ is an isomorphism. Which shows that all cyclic modules are isomorphic to a quotient of R.

Definition 2.1.2. *An element $m \in M$ is called*

- ***torsion-free*** *if* $\mathrm{Ann}_R(m) = 0$.
- ***torsion*** *if* $\mathrm{Ann}_R(m) \neq 0$.

A module is called torsion-free if all its elements except of zero are torsion-free.

Definition 2.1.3. *The torsion part of M is defined as:*

$$\mathrm{Tor}(M) = \{m \in M \mid \mathrm{Ann}_R(m) \neq 0\}.$$

Lemma 2.1.4. *If R is an integral domain, then the torsion submodule of M is a submodule of M.*

Proof. We verify the submodule criteria.

Non-empty. Since $r \cdot 0_M = 0_M$ for every $r \in R$ and $R \neq 0$, we have $\mathrm{Ann}_R(0_M) = R \neq 0$, so $0_M \in \mathrm{Tor}(M)$.

Closed under addition. Let $m, n \in \mathrm{Tor}(M)$. Then there exist $\lambda, \mu \in R \setminus \{0\}$ such that

$\lambda m = 0$ and $\mu n = 0$. Hence

$$(\lambda\mu)(m+n) = \mu(\lambda m) + \lambda(\mu n) = 0.$$

Since R is an integral domain and $\lambda, \mu \neq 0$, we have $\lambda\mu \neq 0$, so $\mathrm{Ann}_R(m+n) \neq 0$ and $m + n \in \mathrm{Tor}(M)$.

Closed under scalar multiplication. Let $m \in \mathrm{Tor}(M)$ and $r \in R$. Choose $\lambda \in R \setminus \{0\}$ with $\lambda m = 0$. Then

$$\lambda(rm) = r(\lambda m) = 0,$$

so $\mathrm{Ann}_R(rm) \neq 0$ and $rm \in \mathrm{Tor}(M)$.

Therefore $\mathrm{Tor}(M)$ is a submodule of M. □

Example 2.1.5. *(1) $0 \in M$ is torsion.*
(2) Let R be a commutative ring with 1. Then R is an integral domain if and only if ${}_RR$ is torsion-free. In this case $\mathrm{Tor}({}_RR) = 0$, $\mathrm{Tor}(R^n) = 0$ for all $n \geq 1$.
(3) If I is a nonzero ideal of R, then $\mathrm{Tor}(R/I) = R/I$ where R/I is regarded as an R-module. If we regard R/I as an R/I-module, then it is torsion-free if and only if I is a prime ideal.
(4) More generally, let I be any non-zero ideal of R, and M be an R-module. Then $\mathrm{Tor}(M/IM) = M/IM$ where M/IM is regarded as an R-module.

Remark 2.1.6. *Let R be a commutative ring with 1. Let M be an R-module. Then $f : M \to M$ such that $f(m) = rm$ for all $m \in M$ and $r \in R$ is a homomorphism of R-modules. If moreover M is torsion-free, then f is a monomorphism.*

Lemma 2.1.7. *If R is an integral domain and M is an R-module, then*

1. $\mathrm{Tor}(\mathrm{Tor}(M)) = \mathrm{Tor}(M)$.

2. $\mathrm{Tor}(M/\mathrm{Tor}(M)) = 0$.

Proof.

1. Since $\mathrm{Tor}(M)$ is a submodule of M, its torsion submodule is

$$\mathrm{Tor}(\mathrm{Tor}(M)) = \{\, m \in \mathrm{Tor}(M) \mid \mathrm{Ann}_R(m) \neq 0 \,\}.$$

Every $m \in \mathrm{Tor}(M)$ already satisfies $\mathrm{Ann}_R(m) \neq 0$, so $\mathrm{Tor}(\mathrm{Tor}(M)) \subseteq \mathrm{Tor}(M)$. Conversely, $\mathrm{Tor}(M) \subseteq \mathrm{Tor}(\mathrm{Tor}(M))$ because the annihilator of m in R does not depend on the ambient module: for any $m \in \mathrm{Tor}(M)$, its annihilator computed in the submodule $\mathrm{Tor}(M)$ is the same as $\mathrm{Ann}_R(m) \neq 0$, so $m \in \mathrm{Tor}(\mathrm{Tor}(M))$. Hence $\mathrm{Tor}(\mathrm{Tor}(M)) = \mathrm{Tor}(M)$.

2. Let $\pi\colon M \to M/\mathrm{Tor}(M)$ be the canonical projection. Write $\overline{m} = \pi(m) = m +$

$\operatorname{Tor}(M)$. Suppose $\overline{m} \in \operatorname{Tor}(M/\operatorname{Tor}(M))$, i.e. there exists $\lambda \in R \setminus \{0\}$ such that

$$\lambda\,\overline{m} = \overline{0} \quad \text{in } M/\operatorname{Tor}(M).$$

This means $\lambda m \in \operatorname{Tor}(M)$, so there exists $\mu \in R \setminus \{0\}$ with $\mu(\lambda m) = 0$, i.e.

$$(\mu\lambda)\,m = 0.$$

Since R is an integral domain and $\mu, \lambda \neq 0$, we have $\mu\lambda \neq 0$, hence $m \in \operatorname{Tor}(M)$, so $\overline{m} = \overline{0}$. Therefore $\operatorname{Tor}(M/\operatorname{Tor}(M)) = 0$.

□

Definition 2.1.8. *Let J be a subscript set, denote $R^J = \prod_{j\in J} R$. $R^{(J)} = \bigoplus_{j\in J} R$. Let M be an R-module. Let $x_j \in M$ where $j \in J$ be a family of elements of M. Then the submodule generated by $(x_j)_{j\in J}$ is denoted by $\sum_{j\in J} Rx_j$.*

Corollary 2.1.9. *We have a morphism of R-modules $\phi : R^{(J)} \to M$ such that $\phi(r_j)_{j\in J} = \sum_{j\in J} r_j x_j$ for all $(r_j)_{j\in J} \in R^{(J)}$.*

- *ϕ is injective if and only if $\{x_j\}_{j\in J}$ are linearly independent.*
- *ϕ is surjective if and only if $\{x_j\}_{j\in J}$ generate M.*
- *ϕ is an isomorphism if and only if M is free with basis $\{x_j\}_{j\in J}$.*

Lemma 2.1.10. *Let L be a free R-module with basis $\{x_j\}_{j\in J}$. For any R-module M the following map:*

$$\operatorname{Hom}_R(L, M) \to M^J, \quad f \mapsto (f(x_j))_{j\in J}$$

is a bijection.

Proposition 2.1.11. *Let L be a free R-module.*

1. *Let $\pi : X \to Y$ be an epimorphism of R-modules. For any R-module homomorphism $\varphi : L \to Y$ there exists a morphism $\tilde{\varphi} : L \to X$ such that the following diagram commutes:*

$$\begin{tikzcd} & L \arrow[dl, "\tilde{\varphi}"'] \arrow[d, "\varphi"] & \\ X \arrow[r, "\pi"'] & Y \arrow[r] & 0 \end{tikzcd}$$

2. *In particular, for any R-module M any epimorphism $\pi : M \to L$ has a section, that is, there exists a morphism $\sigma : L \to M$ such that $\pi\sigma = \mathrm{id}_L$. In that case σ is injective, $\mathrm{im}(\sigma) = L$ and $M = \ker\pi \oplus \mathrm{im}(\sigma)$.*

$$0 \longrightarrow \ker \pi \longrightarrow M \underset{\sigma}{\overset{\longrightarrow\!\!\!\!\rightarrow}{\longleftarrow}} L \longrightarrow 0$$

Proof. Let $\{e_j\}_{j\in J}$ be a basis of L.

(1) Since π is surjective, for each $j \in J$ there exists $\tilde{x}_j \in X$ such that $\pi(\tilde{x}_j) = \varphi(e_j)$. By the lemma above, the map $\mathrm{Hom}_R(L, X) \to X^J$ given by $f \mapsto (f(e_j))_{j\in J}$ is a bijection. Hence there exists a unique R-module homomorphism $\tilde{\varphi} : L \to X$ such that $\tilde{\varphi}(e_j) = \tilde{x}_j$ for all $j \in J$. Then for each basis element e_j,

$$\pi \circ \tilde{\varphi}(e_j) = \pi(\tilde{x}_j) = \varphi(e_j),$$

so $\pi\circ\tilde{\varphi}$ and φ agree on the basis $\{e_j\}_{j\in J}$. Since the basis generates L, we conclude $\pi\circ\tilde{\varphi} = \varphi$.

Remark 2.1.12. *The uniqueness appearing in the proof of the lifting property is* not *the uniqueness of the lift among all morphisms $\tilde{\varphi} : L \to X$ satisfying*

$$\pi \circ \tilde{\varphi} = \varphi.$$

Rather, it is only the uniqueness after a choice of lifts of the basis elements has been fixed.

Indeed, let L be a free R-module with basis $\{e_j\}_{j\in J}$. Since $\pi : X \to Y$ is surjective, for each $j \in J$ we may choose an element $\tilde{x}_j \in X$ such that

$$\pi(\tilde{x}_j) = \varphi(e_j).$$

By the universal property of a free module, or equivalently by the bijection

$$\mathrm{Hom}_R(L, X) \xrightarrow{\sim} X^J, \qquad f \mapsto (f(e_j))_{j\in J},$$

there exists a unique *R-module homomorphism*

$$\tilde{\varphi} : L \to X$$

such that

$$\tilde{\varphi}(e_j) = \tilde{x}_j \qquad \textit{for all } j \in J.$$

Thus the word "unique" here means: for the chosen family $(\tilde{x}_j)_{j\in J}$, there is a unique homomorphism realizing those values on the basis.

However, the elements $\tilde{x}_j$ themselves are generally not *unique. If $\tilde{x}_j$ is one lift of $\varphi(e_j)$, then for any $k_j \in \ker \pi$, the element*

$$\tilde{x}_j + k_j$$

is also a lift, since

$$\pi(\tilde{x}_j + k_j) = \pi(\tilde{x}_j) + \pi(k_j) = \varphi(e_j).$$

Different choices of lifts of the basis elements therefore give, in general, different morphisms $\tilde{\varphi} : L \to X$. Hence the proposition asserts only existence *of a lift, not uniqueness.*

Equivalently, if $\tilde{\varphi}_1, \tilde{\varphi}_2 : L \to X$ are two lifts of φ, then

$$\pi \circ \tilde{\varphi}_1 = \varphi = \pi \circ \tilde{\varphi}_2,$$

so

$$\pi \circ (\tilde{\varphi}_1 - \tilde{\varphi}_2) = 0.$$

Therefore

$$\mathrm{im}(\tilde{\varphi}_1 - \tilde{\varphi}_2) \subseteq \ker \pi,$$

that is,

$$\tilde{\varphi}_1 - \tilde{\varphi}_2 \in \mathrm{Hom}_R(L, \ker \pi).$$

Conversely, if $\delta \in \mathrm{Hom}_R(L, \ker \pi)$, then

$$\pi \circ (\tilde{\varphi}_1 + \delta) = \pi \circ \tilde{\varphi}_1 = \varphi,$$

so $\tilde{\varphi}_1 + \delta$ is again a lift of φ. Thus the lift is unique if and only if

$$\mathrm{Hom}_R(L, \ker \pi) = 0.$$

(2) Apply (1) with $X = M$, $Y = L$, and $\varphi = \mathrm{id}_L$. Then there exists $\sigma : L \to M$ such that $\pi \circ \sigma = \mathrm{id}_L$.

σ is injective: If $\sigma(l) = 0$, then $l = \mathrm{id}_L(l) = \pi(\sigma(l)) = \pi(0) = 0$.

$M = \ker \pi \oplus \mathrm{im}(\sigma)$: For any $m \in M$, write

$$m = (m - \sigma(\pi(m))) + \sigma(\pi(m)).$$

The second summand $\sigma(\pi(m)) \in \mathrm{im}(\sigma)$. At first, $\pi\big(m - \sigma(\pi(m))\big) = \pi(m) - \pi\sigma(\pi(m)) = \pi(m) - \pi(m) = 0$, so it lies in $\ker \pi$. This shows $M = \ker \pi + \mathrm{im}(\sigma)$.

For the directness, if $m \in \ker \pi \cap \mathrm{im}(\sigma)$, write $m = \sigma(l)$ for some $l \in L$. Then $0 = \pi(m) = \pi(\sigma(l)) = l$, so $m = \sigma(0) = 0$. Hence the sum is direct. □

Exercise 2.1.13. *Let L be a free R-module. P a direct summand of L. Show that whenever $\pi : X \to Y$ is an epimorphism and $\varphi : P \to Y$ is a morphism, there exists a morphism $\tilde{\varphi} : P \to X$ such that $\varphi = \pi \circ \tilde{\varphi}$.*

2.2 Tensor product

Let R be a commutative ring. M, N be R-modules. The tensor product $M \otimes_R N$ is a "linearize" of the bilinear map.

Definition 2.2.1. *Let R be a commutative ring. M, N, K be R-modules. $f : M \times N \to K$ is a bilinear map if for all $r \in R$, $m, m_1, m_2 \in M$, $n, n_1, n_2 \in N$, we have:*

- $f(m_1 + m_2, n) = f(m_1, n) + f(m_2, n)$

- $f(m, n_1 + n_2) = f(m, n_1) + f(m, n_2)$
- $f(rm, n) = rf(m, n)$
- $f(m, rn) = rf(m, n)$

Example 2.2.2. *If $f : M \times N \to K$ is a bilinear map, then $f(x, 0) = 0 = f(0, y)$ for all $x \in M$, $y \in N$. $f(-x, y) = -f(x, y)$ and $f(x, -y) = -f(x, y)$ for all $x \in M$, $y \in N$.*

Definition 2.2.3. *The tensor product of M and N is a pair $(M \otimes_R N, t_{M,N})$ where $M \otimes_R N$ is an R-module and $t_{M,N} : M \times N \to M \otimes_R N$ is a bilinear map usually denoted by $t_{M,N}(m, n) = m \otimes n$ which satisfies the following universal properties: Whenever X is an R-module and $f : M \times N \to X$ is a bilinear map, there exists a unique R-module homomorphism $\tilde{f} : M \otimes_R N \to X$ such that the following diagram commutes:*

$$\begin{tikzcd} M \times N \arrow[r, "f"] \arrow[d, "t_{M,N}"'] & X \\ M \otimes_R N \arrow[ur, dashed, "\tilde{f}"'] & \end{tikzcd}$$

Theorem 2.2.4. *The tensor product of M and N exists and is unique up to isomorphism.*

Proof. We prove the uniqueness at first.

The tensor product is unique up to isomorphism. Suppose (T_i, t_1) and (T_2, t_2) are two tensor products of M and N. By the universal property, there exists a unique R-module homomorphism $\varphi : T_1 \to T_2$ such that $t_2 = \varphi \circ t_1$ and a unique R-module homomorphism $\psi : T_2 \to T_1$ such that $t_1 = \psi \circ t_2$. Then we have:

$$\varphi \circ \psi \circ t_2 = \varphi \circ t_1 = t_2$$

$$\psi \circ \varphi \circ t_1 = \psi \circ t_2 = t_1.$$

Since such $\varphi \circ \psi$ and $\psi \circ \varphi$ are unique. It forces $\varphi \circ \psi = \mathrm{id}_{T_2}$ and $\psi \circ \varphi = \mathrm{id}_{T_1}$. Hence φ and ψ are isomorphisms. Therefore $T_1 \cong T_2$.

Now we prove the existence.

Let F be a free R-module with basis $M \times N$. i.e. $F = \mathrm{span}_R(e_{m_i,n_i})_{m_i \in M, n_i \in N}$. Let B be the submodule of F generated by the elements of the following forms:

- $e_{m_1+m_2,n} - e_{m_1,n} - e_{m_2,n}$
- $e_{m,n_1+n_2} - e_{m,n_1} - e_{m,n_2}$
- $e_{rm,n} - re_{m,n}$

- $e_{m,rn} - re_{m,n}$

Where $m, m_1, m_2 \in M$, $n, n_1, n_2 \in N$, $r \in R$. Define $M \otimes_R N = F/B$, $t_{M,N} : M \times N \to M \otimes_R N$ by $t_{M,N}(m,n) = e_{m,n} + B$. It is easy to check that $t_{M,N}$ is a well-defined bilinear map.

We only need to show that $(M \otimes_R N, t_{M,N})$ satisfies the universal property of tensor product:

Let X be an R-module and $f : M \times N \to X$ a bilinear map. Since F is a free R-module with basis $\{e_{m,n}\}_{(m,n)\in M\times N}$, by the universal property of the free module, there exists a unique R-module homomorphism $\hat{f} : F \to X$ such that $\hat{f}(e_{m,n}) = f(m,n)$ for all $(m,n) \in M \times N$.

We claim that $B \subseteq \ker \hat{f}$. Indeed, since f is bilinear:

$$\begin{aligned}
\hat{f}(e_{m_1+m_2,n} - e_{m_1,n} - e_{m_2,n}) &= f(m_1+m_2,n) - f(m_1,n) - f(m_2,n) = 0,\\
\hat{f}(e_{m,n_1+n_2} - e_{m,n_1} - e_{m,n_2}) &= f(m,n_1+n_2) - f(m,n_1) - f(m,n_2) = 0,\\
\hat{f}(e_{rm,n} - re_{m,n}) &= f(rm,n) - rf(m,n) = 0,\\
\hat{f}(e_{m,rn} - re_{m,n}) &= f(m,rn) - rf(m,n) = 0.
\end{aligned}$$

Since B is generated by elements of the above forms and $\ker \hat{f}$ is a submodule, we have $B \subseteq \ker \hat{f}$.

By the universal property of quotient modules, $\hat{f}$ induces a unique R-module homomorphism $\tilde{f} : M \otimes_R N = F/B \to X$ such that $\tilde{f} \circ \pi = \hat{f}$, where $\pi : F \to F/B$ is the canonical projection. In particular, $\tilde{f}(m \otimes n) = \tilde{f}(e_{m,n} + B) = \hat{f}(e_{m,n}) = f(m,n)$, so $\tilde{f} \circ t_{M,N} = f$.

For uniqueness, suppose $g : M \otimes_R N \to X$ is another R-module homomorphism with $g \circ t_{M,N} = f$. Then $g(m \otimes n) = f(m,n) = \tilde{f}(m \otimes n)$ for all $m \in M, n \in N$. Since elements of the form $m \otimes n$ generate $M \otimes_R N$ as an R-module, $g = \tilde{f}$.

Therefore $(M \otimes_R N, t_{M,N})$ is a tensor product of M and N. $\square$

Definition 2.2.5. *The set of elements of the form $m \otimes n$ for $m \in M$ and $n \in N$ is called the set of elementary tensors. This set generates $M \otimes_R N$ as an R-module.*

Example 2.2.6. *Fix $m \in M$. The map $N \to M \otimes_R N, n \mapsto m \otimes n$ is a morphism of R-modules. Its image $m \otimes_R N = \{m \otimes n \mid n \in N\}$ is a submodule of $M \otimes_R N$ which is isomorphic to a quotient $N/\operatorname{Ann}_R(m)N$.*

Proposition 2.2.7. *We have the following isomorphisms:*

1. $(M_1 \otimes_R M_2) \otimes_R M_3 \cong M_1 \otimes_R (M_2 \otimes_R M_3)$ *where* $(m_1 \otimes m_2) \otimes m_3 \mapsto m_1 \otimes (m_2 \otimes m_3)$.
2. $M_1 \otimes_R M_2 \cong M_2 \otimes_R M_1$ *where* $m_1 \otimes m_2 \mapsto m_2 \otimes m_1$.
3. $M_1 \otimes_R (M_2 \oplus M_3) \cong (M_1 \otimes_R M_2) \oplus (M_1 \otimes_R M_3)$ *where* $m_1 \otimes (m_2 \oplus m_3) \mapsto$

$(m_1 \otimes m_2) + (m_1 \otimes m_3)$. *And more generally,*

$$M_1 \otimes_R (\bigoplus_{i=1}^{n} M_i) \cong \bigoplus_{i=1}^{n} (M_1 \otimes_R M_i)$$

where $m_1 \otimes (m_2 \oplus \cdots \oplus m_n) \mapsto (m_1 \otimes m_2) + \cdots + (m_1 \otimes m_n)$.

4. $R \otimes_R M \cong M$ *where* $r \otimes m \mapsto rm$. *And more generally,* $R^{(I)} \otimes_R M \cong M^{(I)}$ *where* $(r_i)_{i\in I} \otimes m \mapsto (r_i m)_{i\in I}$.

Proof. We prove (1): $(M_1 \otimes_R M_2) \otimes_R M_3 \cong M_1 \otimes_R (M_2 \otimes_R M_3)$.

Step 1. We construct $\varphi : (M_1 \otimes_R M_2) \otimes_R M_3 \to M_1 \otimes_R (M_2 \otimes_R M_3)$.

Fix $m_3 \in M_3$. For each m_3, define $f_{m_3} : M_1 \times M_2 \to M_1 \otimes_R (M_2 \otimes_R M_3)$ by $f_{m_3}(m_1, m_2) = m_1 \otimes (m_2 \otimes m_3)$. This is bilinear in (m_1, m_2), so by the universal property of $M_1 \otimes_R M_2$, there exists a unique R-module homomorphism $\tilde{f}_{m_3} : M_1 \otimes_R M_2 \to M_1 \otimes_R (M_2 \otimes_R M_3)$ such that $\tilde{f}_{m_3}(m_1 \otimes m_2) = m_1 \otimes (m_2 \otimes m_3)$:

$$\begin{tikzcd} M_1 \times M_2 \arrow[r, "f_{m_3}"] \arrow[d, "t_{M_1,M_2}"'] & M_1 \otimes_R (M_2 \otimes_R M_3) \\ M_1 \otimes_R M_2 \arrow[ur, dashed, "\tilde{f}_{m_3}"'] & \end{tikzcd}$$

Now define $g : (M_1 \otimes_R M_2) \times M_3 \to M_1 \otimes_R (M_2 \otimes_R M_3)$ by $g(\alpha, m_3) = \tilde{f}_{m_3}(\alpha)$. We verify that g is bilinear. Linearity in the first argument follows because each $\tilde{f}_{m_3}$ is an R-module homomorphism. For the second argument, on elementary tensors:

$$\begin{aligned} g(m_1 \otimes m_2, m_3 + m_3') &= m_1 \otimes (m_2 \otimes (m_3 + m_3')) \\ &= m_1 \otimes (m_2 \otimes m_3 + m_2 \otimes m_3') \\ &= m_1 \otimes (m_2 \otimes m_3) + m_1 \otimes (m_2 \otimes m_3') \\ &= g(m_1 \otimes m_2, m_3) + g(m_1 \otimes m_2, m_3'), \\ g(m_1 \otimes m_2, rm_3) &= m_1 \otimes (m_2 \otimes rm_3) \\ &= m_1 \otimes (r(m_2 \otimes m_3)) \\ &= r(m_1 \otimes (m_2 \otimes m_3)) \\ &= r\, g(m_1 \otimes m_2, m_3). \end{aligned}$$

Since elementary tensors generate $M_1 \otimes_R M_2$, bilinearity holds on all of $(M_1 \otimes_R M_2) \times M_3$.

By the universal property of $(M_1 \otimes_R M_2) \otimes_R M_3$, there exists a unique R-module homomorphism

$$\varphi : (M_1 \otimes_R M_2) \otimes_R M_3 \to M_1 \otimes_R (M_2 \otimes_R M_3)$$

such that $\varphi((m_1 \otimes m_2) \otimes m_3) = m_1 \otimes (m_2 \otimes m_3)$:

$$\begin{tikzcd} (M_1 \otimes_R M_2) \times M_3 \arrow[r, "g"] \arrow[d, "t"'] & M_1 \otimes_R (M_2 \otimes_R M_3) \\ (M_1 \otimes_R M_2) \otimes_R M_3 \arrow[ur, dashed, "\varphi"'] & \end{tikzcd}$$

Step 2. By an entirely symmetric argument (fixing m_1 first, then extending), we obtain an R-module homomorphism

$$\psi : M_1 \otimes_R (M_2 \otimes_R M_3) \to (M_1 \otimes_R M_2) \otimes_R M_3$$

such that $\psi(m_1 \otimes (m_2 \otimes m_3)) = (m_1 \otimes m_2) \otimes m_3$.

Step 3. We show φ and ψ are inverses. On elementary tensors:

$$(\psi \circ \varphi)((m_1 \otimes m_2) \otimes m_3) = \psi(m_1 \otimes (m_2 \otimes m_3)) = (m_1 \otimes m_2) \otimes m_3,$$

$$(\varphi \circ \psi)(m_1 \otimes (m_2 \otimes m_3)) = \varphi((m_1 \otimes m_2) \otimes m_3) = m_1 \otimes (m_2 \otimes m_3).$$

Since elementary tensors generate both modules, $\psi \circ \varphi = \mathrm{id}$ and $\varphi \circ \psi = \mathrm{id}$. The overall situation is summarized by the following commutative diagram:

$$\begin{tikzcd} & M_1 \times M_2 \times M_3 \arrow[dl] \arrow[dr] & \\ (M_1 \otimes_R M_2) \otimes_R M_3 \arrow[rr, shift left, "\varphi"] & & M_1 \otimes_R (M_2 \otimes_R M_3) \arrow[ll, shift left, "\psi"] \end{tikzcd}$$

where the diagonal arrows send (m_1, m_2, m_3) to $(m_1 \otimes m_2) \otimes m_3$ and $m_1 \otimes (m_2 \otimes m_3)$ respectively. Hence φ is an isomorphism with $(m_1 \otimes m_2) \otimes m_3 \mapsto m_1 \otimes (m_2 \otimes m_3)$. □

Theorem 2.2.8. *Let M, N, P be R-modules. We have a canonical isomorphism:*

$$\phi_{M,N,P} : \mathrm{Hom}_R(M \otimes_R N, P) \to \mathrm{Hom}_R(M, \mathrm{Hom}_R(N, P))$$

such that $\phi_{M,N,P}(f)(m)(n) = f(m \otimes n)$. Furthermore, if R is commutative, then $\phi_{M,N,P}$ is an R-module isomorphism. If R is not commutative, then $\phi_{M,N,P}$ is only a $\mathbb{Z}$-module isomorphism.

Proof. Let $f \in \mathrm{Hom}_R(M \otimes_R N, P)$. For each $m \in M$, define $\phi_{M,N,P}(f)(m) : N \to P$ by

$$\phi_{M,N,P}(f)(m)(n) = f(m \otimes n).$$

We verify that $\phi_{M,N,P}(f)(m) \in \mathrm{Hom}_R(N, P)$: for $n_1, n_2 \in N$ and $r \in R$,

$$\begin{aligned} \phi_{M,N,P}(f)(m)(n_1 + n_2) &= f(m \otimes (n_1 + n_2)) = f(m \otimes n_1 + m \otimes n_2) \\ &= f(m \otimes n_1) + f(m \otimes n_2), \\ \phi_{M,N,P}(f)(m)(rn) &= f(m \otimes rn) = f(r(m \otimes n)) \\ &= rf(m \otimes n) = r\,\phi_{M,N,P}(f)(m)(n). \end{aligned}$$

Next, we verify that $\phi_{M,N,P}(f) \in \mathrm{Hom}_R(M, \mathrm{Hom}_R(N, P))$: for $m_1, m_2 \in M$, $r \in R$, and any $n \in N$,

$$\begin{aligned}
\phi_{M,N,P}(f)(m_1 + m_2)(n) &= f((m_1 + m_2) \otimes n) \\
&= f(m_1 \otimes n) + f(m_2 \otimes n) \\
\phi_{M,N,P}(f)(rm)(n) &= f(rm \otimes n) \\
&= f(r(m \otimes n)) \\
&= rf(m \otimes n) \\
&= (r\, \phi_{M,N,P}(f)(m))(n).
\end{aligned}$$

So

$$\phi_{M,N,P}(f)(m_1 + m_2) = \phi_{M,N,P}(f)(m_1) + \phi_{M,N,P}(f)(m_2)$$

and $\phi_{M,N,P}(f)(rm) = r\, \phi_{M,N,P}(f)(m)$, confirming R-linearity.

Let $g \in \mathrm{Hom}_R(M, \mathrm{Hom}_R(N, P))$. Define $\hat{g} : M \times N \to P$ by $\hat{g}(m, n) = g(m)(n)$. We check that $\hat{g}$ is R-bilinear:

$$\begin{aligned}
\hat{g}(m_1 + m_2, n) &= g(m_1 + m_2)(n) = g(m_1)(n) + g(m_2)(n) = \hat{g}(m_1, n) + \hat{g}(m_2, n), \\
\hat{g}(m, n_1 + n_2) &= g(m)(n_1 + n_2) = g(m)(n_1) + g(m)(n_2) = \hat{g}(m, n_1) + \hat{g}(m, n_2), \\
\hat{g}(rm, n) &= g(rm)(n) = (rg(m))(n) = rg(m)(n) = r\, \hat{g}(m, n), \\
\hat{g}(m, rn) &= g(m)(rn) = rg(m)(n) = r\, \hat{g}(m, n).
\end{aligned}$$

By the universal property of $M \otimes_R N$, there exists a unique R-module homomorphism $\psi_{M,N,P}(g) : M \otimes_R N \to P$ such that $\psi_{M,N,P}(g)(m \otimes n) = g(m)(n)$:

$$\begin{tikzcd}
M \times N \arrow[r, "\hat{g}"] \arrow[d, "t_{M,N}"'] & P \\
M \otimes_R N \arrow[ur, dashed, "\psi_{M,N,P}(g)"'] &
\end{tikzcd}$$

For any $f \in \mathrm{Hom}_R(M \otimes_R N, P)$:

$$\psi_{M,N,P}(\phi_{M,N,P}(f))(m \otimes n) = \phi_{M,N,P}(f)(m)(n) = f(m \otimes n).$$

Since elementary tensors generate $M \otimes_R N$, we have $\psi_{M,N,P}(\phi_{M,N,P}(f)) = f$.

For any $g \in \mathrm{Hom}_R(M, \mathrm{Hom}_R(N, P))$, for all $m \in M$ and $n \in N$:

$$\phi_{M,N,P}(\psi_{M,N,P}(g))(m)(n) = \psi_{M,N,P}(g)(m \otimes n) = g(m)(n).$$

So $\phi_{M,N,P}(\psi_{M,N,P}(g))(m) = g(m)$ for all m, hence $\phi_{M,N,P}(\psi_{M,N,P}(g)) = g$.

Therefore $\phi_{M,N,P}$ is a bijection of sets (in fact, of abelian groups).

It remains to show that $\phi_{M,N,P}$ respects the appropriate module structure.

Both $\mathrm{Hom}_R(M \otimes_R N, P)$ and $\mathrm{Hom}_R(M, \mathrm{Hom}_R(N, P))$ are abelian groups under pointwise addition. We verify $\phi_{M,N,P}$ is a group homomorphism: for $f_1, f_2 \in \mathrm{Hom}_R(M \otimes_R N, P)$,

$$\begin{aligned}\phi_{M,N,P}(f_1+f_2)(m)(n) &= (f_1+f_2)(m\otimes n)\\ &= f_1(m\otimes n)+f_2(m\otimes n)\\ &= \phi_{M,N,P}(f_1)(m)(n)+\phi_{M,N,P}(f_2)(m)(n)\end{aligned}$$

so $\phi_{M,N,P}(f_1+f_2) = \phi_{M,N,P}(f_1)+\phi_{M,N,P}(f_2)$. Hence $\phi_{M,N,P}$ is a $\mathbb{Z}$-module isomorphism.

If R is commutative, $\mathrm{Hom}_R(M \otimes_R N, P)$ carries an R-module structure via $(r\cdot f)(x) = rf(x)$, and similarly for $\mathrm{Hom}_R(M, \mathrm{Hom}_R(N,P))$. Then

$$\begin{aligned}\phi_{M,N,P}(rf)(m)(n) &= (rf)(m\otimes n)\\ &= rf(m\otimes n)\\ &= r\,\phi_{M,N,P}(f)(m)(n)\\ &= (r\,\phi_{M,N,P}(f))(m)(n)\end{aligned}$$

so $\phi_{M,N,P}(rf) = r\,\phi_{M,N,P}(f)$, and $\phi_{M,N,P}$ is an R-module isomorphism.

If R is not commutative, $\mathrm{Hom}_R(M \otimes_R N, P)$ does not naturally carry an R-module structure (scalar multiplication by r need not commute with R-linearity), so $\phi_{M,N,P}$ is only a $\mathbb{Z}$-module isomorphism. $\square$

Definition 2.2.9. *Let $f : M \to M'$ and $g : N \to N'$ be R-module homomorphisms. The tensor product of f and g is the R-module homomorphism:*

$$f \otimes g : M \otimes_R N \to M' \otimes_R N'$$

defined by $(f\otimes g)(m\otimes n) = f(m)\otimes g(n)$.

Definition 2.2.10. *Let $M_i, i \in I$ be a family of R-modules. Use the following notation:*

$$L^n(M_1,\dots,M_n;N) = \{f : M_1\times\cdots\times M_n \to N \mid f \text{ is } n\text{-linear }\}$$

In particular,

$$L^2(M_1,M_2;N) \cong L(M_1\otimes M_2, N) \cong \mathrm{Hom}_R(M_1\otimes M_2, N)$$

or equivalently,

$$L^2(M_1\otimes M_2, N) \cong L(M_1, L(M_2,N))$$

Proposition 2.2.11. *Let M, N be two finitely generated free R-modules. Denote*

$$M^\vee = \mathrm{Hom}_R(M,R)$$

Then we have a canonical isomorphism:

$$M^\vee \otimes N \cong \mathrm{Hom}_R(M, N)$$

Proof. Since M is a finitely generated free R-module, we have $M \cong R^m$ for some $m \geq 1$. We construct an explicit isomorphism

$$\Phi : M^\vee \otimes_R N \to \mathrm{Hom}_R(M, N).$$

Define a bilinear map $\beta : M^\vee \times N \to \mathrm{Hom}_R(M, N)$ by

$$\beta(f, n)(m) = f(m)\, n, \qquad f \in M^\vee,\ n \in N,\ m \in M.$$

We check R-bilinearity. For $f_1, f_2 \in M^\vee$, $n, n_1, n_2 \in N$, $m \in M$, and $r \in R$:

$$\begin{aligned}
\beta(f_1 + f_2, n)(m) &= (f_1 + f_2)(m)\, n = f_1(m)\, n + f_2(m)\, n \\
&= \beta(f_1, n)(m) + \beta(f_2, n)(m), \\
\beta(f, n_1 + n_2)(m) &= f(m)(n_1 + n_2) = f(m)\, n_1 + f(m)\, n_2 \\
&= \beta(f, n_1)(m) + \beta(f, n_2)(m), \\
\beta(rf, n)(m) &= (rf)(m)\, n = r\, f(m)\, n = r\, \beta(f, n)(m), \\
\beta(f, rn)(m) &= f(m)\, (rn) = r\, f(m)\, n = r\, \beta(f, n)(m).
\end{aligned}$$

By the universal property of the tensor product, there exists a unique R-module homomorphism

$$\Phi : M^\vee \otimes_R N \to \mathrm{Hom}_R(M, N), \qquad \Phi(f \otimes n)(m) = f(m)\, n.$$

Now let $\{e_1, \ldots, e_m\}$ be a basis of $M \cong R^m$, and let $\{e_1^*, \ldots, e_m^*\}$ be the dual basis, i.e. $e_i^* \in M^\vee$ with $e_i^*(e_j) = \delta_{ij}$. Define

$$\Psi : \mathrm{Hom}_R(M, N) \to M^\vee \otimes_R N, \qquad \Psi(g) = \sum_{i=1}^m e_i^* \otimes g(e_i).$$

We verify Ψ is an R-module homomorphism: for $g_1, g_2 \in \mathrm{Hom}_R(M, N)$ and $r \in R$,

$$\begin{aligned}
\Psi(g_1 + g_2) &= \sum_{i=1}^m e_i^* \otimes (g_1 + g_2)(e_i) = \sum_{i=1}^m e_i^* \otimes (g_1(e_i) + g_2(e_i)) \\
&= \sum_{i=1}^m e_i^* \otimes g_1(e_i) + \sum_{i=1}^m e_i^* \otimes g_2(e_i) = \Psi(g_1) + \Psi(g_2), \\
\Psi(rg) &= \sum_{i=1}^m e_i^* \otimes (rg)(e_i) = \sum_{i=1}^m e_i^* \otimes r\, g(e_i) \\
&= r \sum_{i=1}^m e_i^* \otimes g(e_i) = r\, \Psi(g).
\end{aligned}$$

We show Φ and Ψ are mutual inverses. For any $g \in \mathrm{Hom}_R(M,N)$ and $m = \sum_j r_j e_j \in M$:

$$\begin{aligned}\Phi(\Psi(g))(m) &= \Phi\Big(\sum_{i=1}^{m} e_i^* \otimes g(e_i)\Big)\Big(\sum_j r_j e_j\Big)\\ &= \sum_{i=1}^{m} e_i^*\Big(\sum_j r_j e_j\Big) g(e_i) = \sum_{i=1}^{m} r_i\, g(e_i)\\ &= g\Big(\sum_{i=1}^{m} r_i e_i\Big) = g(m).\end{aligned}$$

So $\Phi \circ \Psi = \mathrm{id}$.

For any elementary tensor $f \otimes n \in M^\vee \otimes_R N$:

$$\begin{aligned}\Psi(\Phi(f\otimes n)) &= \sum_{i=1}^{m} e_i^* \otimes \Phi(f\otimes n)(e_i) = \sum_{i=1}^{m} e_i^* \otimes f(e_i)\, n\\ &= \sum_{i=1}^{m} f(e_i)\,(e_i^* \otimes n) = \Big(\sum_{i=1}^{m} f(e_i)\, e_i^*\Big) \otimes n = f \otimes n,\end{aligned}$$

where the last equality uses $f = \sum_{i=1}^{m} f(e_i)\, e_i^*$ (expansion of f in the dual basis). Since elementary tensors generate $M^\vee \otimes_R N$ and $\Psi \circ \Phi$ is R-linear, we have $\Psi \circ \Phi = \mathrm{id}$.

Therefore Φ is an R-module isomorphism, giving $M^\vee \otimes_R N \cong \mathrm{Hom}_R(M,N)$. $\square$

Proposition 2.2.12. *Assume that R is a subring of S, and M is an R-module. Then the bilinear map:*

$$S \times (S \otimes_R M) \to S \otimes_R M$$

defined by

$$(s, s' \otimes m) \mapsto ss' \otimes m$$

defines a structure of S-module on the R-module $S \otimes_R M$.
More generally, let $f : R \to T$ be a ring homomorphism. And view T as an R-module via $RxT \to T, (\lambda, \mu) \mapsto f(\lambda)\mu$. Then the bilinear map:

$$T \times (T \otimes_R M) \to T \otimes_R M$$

defined by

$$(t, t' \otimes m) \mapsto tt' \otimes m$$

defines a structure of T-module on the R-module $T \otimes_R M$.

Proposition 2.2.13. *Let I be an ideal of R. Let M be an R-module. Then the morphism of R-modules:*

$$M \to (R/I) \otimes_R M$$

defined by

$$m \mapsto 1_{R/I} \otimes m$$

induces an isomorphism of R-modules:

$$M/(IM) \cong (R/I) \otimes_R M$$

as R/I-modules.

2.3 Balanced product

Definition 2.3.1. *Let R be a ring (not necessarily commutative). Let M_R be a right R-module and ${}_RN$ be a left R-module. The balanced product of M and N is defined as a pair (P, f) where P is an abelian group and $f : M \times N \to P$ is a bilinear map satisfies:*

- $f(rm, n) = f(m, rn)$ *for all* $r \in R$, $m \in M$, $n \in N$.
- $f(m + m', n) = f(m, n) + f(m', n)$ *for all* $m, m' \in M$, $n \in N$.
- $f(m, n + n') = f(m, n) + f(m, n')$ *for all* $m \in M$, $n, n' \in N$.

Definition 2.3.2. *A tensor product of M_R and ${}_RN$ is a balanced product $(M \otimes_R N, t_{M,N})$ such that if (P, f) is any balanced product of M_R and ${}_RN$, then there exists a unique abelian group homomorphism $\tilde{f} : M \otimes_R N \to P$ such that the following diagram commutes:*

$$\begin{tikzcd} M \times N \arrow[r, "t_{M,N}"] \arrow[d, "f"'] & M \otimes_R N \arrow[dl, dashed, "\tilde{f}"] \\ P & \end{tikzcd}$$

Chapter 3

Lecture 3: Categories, Bimodules

3.1 Categories

Definition 3.1.1. *A category $\mathcal{C}$ consists of:*

- *A class of objects* $\mathrm{ob}(\mathcal{C})$.
- *A set of morphisms* $\mathrm{Hom}_{\mathcal{C}}(A, B)$ *for each pair of objects* $A, B \in \mathcal{C}$.
- *A composition map* $\mathrm{Hom}_{\mathcal{C}}(B, C) \times \mathrm{Hom}_{\mathcal{C}}(A, B) \to \mathrm{Hom}_{\mathcal{C}}(A, C)$ *such that* $(\phi, \psi) \mapsto \phi \circ \psi$ *for each triple of objects* $A, B, C \in \mathcal{C}$.

It is assumed that objects and morphisms satisfy the following conditions:

- If $(A, B) \neq (C, D)$, then
$$\mathrm{Hom}(A, B) \cap \mathrm{Hom}(C, D) = \varnothing.$$
- If
$$f \in \mathrm{Hom}(A, B), \qquad g \in \mathrm{Hom}(B, C), \qquad h \in \mathrm{Hom}(C, D),$$
then
$$(h \circ g) \circ f = h \circ (g \circ f).$$
- (Unit) For every object A we have an element
$$1_A \in \mathrm{Hom}(A, A)$$
such that
$$f \circ 1_A = f \qquad \text{for any } f \in \mathrm{Hom}(A, B),$$
and
$$1_A \circ g = g \qquad \text{for any } g \in \mathrm{Hom}(B, A).$$

Example 3.1.2. *The following are examples of categories.*

1. *The category of sets: the objects are sets, and the morphisms are maps. It is denoted by* **Set**.

2. *The category of groups: the objects are groups, and the morphisms are group homomorphisms. It is denoted by* **Grp**.

3. *The category of abelian groups: the objects are abelian groups, and the morphisms are group homomorphisms. It is denoted by* **Ab**.

4. *The category of R-modules: the objects are R-modules, and the morphisms are R-module homomorphisms. It is denoted by R-***Mod** *(left R-modules, for right R-modules, we use R-***Mod***).*

5. *The category of finitely generated R-modules: the objects are finitely generated R-modules, and the morphisms are R-module homomorphisms. It is denoted by R-***mod**.

6. *The category of topological spaces: the objects are topological spaces, and the morphisms are continuous maps. It is denoted by* **Top**.

Definition 3.1.3 (Subcategory)**.** *A category $\mathcal{D}$ is called a* subcategory *of a category $\mathcal{C}$ if*

1. *the objects of $\mathcal{D}$ form a subclass of the objects of $\mathcal{C}$;*

2. *for any objects A, B of $\mathcal{D}$, we have*

   - $\mathrm{Hom}_{\mathcal{D}}(A, B) \subseteq \mathrm{Hom}_{\mathcal{C}}(A, B)$*;*
   - *the identity morphisms and the composition of morphisms in $\mathcal{D}$ are the same as those in $\mathcal{C}$.*

The subcategory $\mathcal{D}$ of $\mathcal{C}$ is called full *if*

$$\mathrm{Hom}_{\mathcal{D}}(A, B) = \mathrm{Hom}_{\mathcal{C}}(A, B)$$

for any objects A, B of $\mathcal{D}$.

Example 3.1.4. *The category* **Ab** *is a full subcategory of* **Grp**.
*The category R-***mod** *is a full subcategory of R-***Mod**.
The category **Grp** *is not a full subcategory of* **Set**.

Definition 3.1.5. *Let $\mathcal{C}$ and $\mathcal{D}$ be categories. A* covariant functor

$$F : \mathcal{C} \to \mathcal{D}$$

consists of the following data:

- *for every object A of $\mathcal{C}$, an object $F(A)$ of $\mathcal{D}$;*
- *for each pair of objects A, B of $\mathcal{C}$, a map*

$$F : \mathrm{Hom}_{\mathcal{C}}(A, B) \to \mathrm{Hom}_{\mathcal{D}}(F(A), F(B)), \qquad f \mapsto F(f).$$

These data are required to satisfy the following conditions:

1. *If $g \circ f$ is defined in $\mathcal{C}$, then*

$$F(g) \circ F(f) = F(g \circ f).$$

2. *For every object A of $\mathcal{C}$,*

$$F(1_A) = 1_{F(A)}.$$

Proposition 3.1.6. *Functors preserve diagram:*

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[dr, "g"'] & B \arrow[d, "h"] \\ & C \end{tikzcd} \quad \xrightarrow{F} \quad \begin{tikzcd} F(A) \arrow[r, "F(f)"] \arrow[dr, "F(g)"'] & F(B) \arrow[d, "F(h)"] \\ & F(AC \end{tikzcd}$$

Let M and N be R-modules. Then

$$\mathrm{Hom}_R(M, N) = \{\, f : M \to N \mid f \text{ is an } R\text{-module homomorphism} \,\}.$$

This is an abelian group.

Moreover,

$$\mathrm{End}_R(M) = \mathrm{Hom}_R(M, M),$$

and $\mathrm{End}_R(M)$ is a ring.

Example 3.1.7. *Let M be an R-module. Then*

$$\mathrm{Hom}_R(M, -) : R\text{-Mod} \to \mathrm{Ab}$$

is a functor.
More precisely, it acts as follows:

$$N \longmapsto \mathrm{Hom}_R(M, N), \qquad f : N \to N' \longmapsto \mathrm{Hom}_R(M, f).$$

Given an R-module homomorphism

$$f : N \to N',$$

the induced map

$$\mathrm{Hom}_R(M,f) : \mathrm{Hom}_R(M,N) \to \mathrm{Hom}_R(M,N')$$

is defined by

$$\mathrm{Hom}_R(M,f)(h) = f \circ h \qquad (h \in \mathrm{Hom}_R(M,N)).$$

This corresponds to the commutative diagram

$$\begin{tikzcd} & M \arrow[dl, "h"'] \arrow[dr, "f\circ h"] & \\ N \arrow[rr, "f"'] & & N' \end{tikzcd}$$

Hence

$$\mathrm{Hom}_R(M,-)$$

is a covariant functor.

Definition 3.1.8. *Let $\mathcal{C}$ be a category. The* opposite category *(or* dual category*) $\mathcal{C}^{\mathrm{op}}$ is defined as follows:*

- $\mathrm{Ob}(\mathcal{C}^{\mathrm{op}}) = \mathrm{Ob}(\mathcal{C})$*;*
- *for any objects A, B,*

$$\mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(A,B) = \mathrm{Hom}_{\mathcal{C}}(B,A).$$

The composition in $\mathcal{C}^{\mathrm{op}}$ is defined by reversing the order of composition in $\mathcal{C}$: if

$$f \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(A,B), \qquad g \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(B,C),$$

then

$$g \circ f \text{ in } \mathcal{C}^{\mathrm{op}}$$

is defined to be

$$f \circ g \text{ in } \mathcal{C}.$$

We call $\mathcal{C}^{\mathrm{op}}$ the dual category of $\mathcal{C}$.

This is illustrated by the following diagrams:

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[dr, "g\circ f"'] & B \arrow[d, "g"] \\ & C \end{tikzcd} \qquad \text{in } \mathcal{C}^{\mathrm{op}},$$

while the corresponding picture in $\mathcal{C}$ is

$$\begin{tikzcd} A & B \arrow[l, "f"'] \\ & C \arrow[u, "g"'] \arrow[ul, "f\circ g"] \end{tikzcd} \qquad \text{in } \mathcal{C}.$$

Definition 3.1.9. *A* contravariant functor *from a category $\mathcal{C}$ to a category $\mathcal{D}$ is a covariant functor*

$$\mathcal{C}^{\mathrm{op}} \to \mathcal{D}.$$

Example 3.1.10. *Let M be an R-module. Then*

$$\mathrm{Hom}_R(-, M) : R\text{-Mod} \to \mathrm{Ab}$$

is a contravariant functor.

More precisely, it is defined on objects by

$$N \longmapsto \mathrm{Hom}_R(N, M),$$

and on morphisms by sending an R-module homomorphism

$$f : N \to N'$$

to the induced group homomorphism

$$\mathrm{Hom}_R(f, M) : \mathrm{Hom}_R(N', M) \to \mathrm{Hom}_R(N, M),$$

given by

$$\mathrm{Hom}_R(f, M)(h) = h \circ f \qquad (h \in \mathrm{Hom}_R(N', M)).$$

This corresponds to the commutative diagram

$$\begin{tikzcd} N \arrow[d, "f"'] \arrow[dr, "h\circ f"] & \\ N' \arrow[r, "h"'] & M \end{tikzcd}$$

3.2 Exact Sequences

Definition 3.2.1. *Let $\{M_i\}_{i\in\mathbb{Z}}$ be a family of R-modules, and let*

$$\varphi_i : M_i \to M_{i+1}$$

be R-module homomorphisms for each $i \in \mathbb{Z}$. Then

$$\cdots \longrightarrow M_{i-1} \xrightarrow{\varphi_{i-1}} M_i \xrightarrow{\varphi_i} M_{i+1} \longrightarrow \cdots$$

is called a sequence *of R-modules.*

Definition 3.2.2. *The sequence*

$$\cdots \longrightarrow M_{i-1} \xrightarrow{\varphi_{i-1}} M_i \xrightarrow{\varphi_i} M_{i+1} \longrightarrow \cdots$$

is said to be exact at M_i *if*

$$\operatorname{Im}(\varphi_{i-1}) = \ker(\varphi_i).$$

It is called an exact sequence *if it is exact at every term.*

A short exact sequence is a sequence of the form

$$0 \longrightarrow M' \xrightarrow{\iota} M \xrightarrow{\pi} M'' \longrightarrow 0.$$

It is exact if and only if

$$\ker(\iota) = 0, \qquad \operatorname{Im}(\pi) = M'', \qquad \operatorname{Im}(\iota) = \ker(\pi).$$

Equivalently, ι is injective, π is surjective, and

$$\operatorname{Im}(\iota) = \ker(\pi).$$

Hence, for every short exact sequence

$$0 \longrightarrow M' \xrightarrow{\iota} M \xrightarrow{\pi} M'' \longrightarrow 0,$$

we have

$$M/M' \cong M''.$$

Definition 3.2.3. *Let R, S be rings, and let*

$$F : R\text{-Mod} \to S\text{-Mod}$$

be a covariant functor.
We say that F is left exact *if for any exact sequence*

$$0 \to N \xrightarrow{\varphi} M \xrightarrow{\psi} P,$$

the sequence

$$0 \to F(N) \xrightarrow{F(\varphi)} F(M) \xrightarrow{F(\psi)} F(P)$$

is exact.
We say that F is right exact *if for any exact sequence*

$$N \xrightarrow{\varphi} M \xrightarrow{\psi} P \to 0,$$

the sequence

$$F(N) \xrightarrow{F(\varphi)} F(M) \xrightarrow{F(\psi)} F(P) \to 0$$

is exact.

Definition 3.2.4. *Let*

$$G : R\text{-Mod} \to S\text{-Mod}$$

be a contravariant functor. We say that G is left exact *(respectively,* right exact*) if it is left exact (respectively, right exact) when regarded as a covariant functor*

$$G : (R\text{-Mod})^{\mathrm{op}} \to S\text{-Mod}.$$

A functor is called exact *if it is both left exact and right exact.*

Let M be an R-module. Then:

$$\mathrm{Hom}_R(M, -) : R\text{-Mod} \to \mathrm{Ab} \quad \text{is left exact,}$$

$$\mathrm{Hom}_R(-, M) : R\text{-Mod} \to \mathrm{Ab} \quad \text{is right exact,}$$

and

$$M \otimes_R - : R\text{-Mod} \to \mathrm{Ab} \quad \text{is right exact.}$$

Definition 3.2.5. *A functor $F : \mathcal{C} \to \mathcal{D}$ is called* exact *if it is both left exact and right exact.*

Definition 3.2.6. *Let M be an R-module.*

1. *M is called* projective *if the functor*

$$\mathrm{Hom}_R(M, -)$$

   is exact.

2. *M is called* injective *if the functor*

$$\mathrm{Hom}_R(-, M)$$

   is exact.

3. *M is called* flat *if the functor*

$$M \otimes_R -$$

is exact.

If R is noncommutative, then in order for

$$M \otimes_R -$$

to be well-defined, M must be a right R-module.

3.3 Bimodules

Definition 3.3.1. *Let R and S be rings. An abelian group M is called a* left-R-right-S-bimodule *if M is both a left R-module and a right S-module such that the two scalar multiplications satisfy*

$$r(xs) = (rx)s \qquad \text{for all } r \in R,\ s \in S,\ x \in M.$$

Notation.

$$_RM \qquad \text{left } R\text{-module},$$

$$M_S \qquad \text{right } S\text{-module},$$

$$_RM_S \qquad \text{left } R\text{-right } S\text{-bimodule}.$$

Let R, S be rings, and let $_RU_S$ be a bimodule.

For each left R-module $_RM$, there are two naturally induced S-module structures:

$$\mathrm{Hom}_R(_RU_S, {}_RM), \qquad \text{a left } S\text{-module},$$

$$\mathrm{Hom}_R(_RM, {}_RU_S), \qquad \text{a right } S\text{-module}.$$

More precisely, for each $s \in S$:

if

$$f \in \mathrm{Hom}_R(_RU_S, {}_RM),$$

define

$$s \cdot f : U \to M, \qquad u \mapsto f(us).$$

Similarly, if

$$f \in \mathrm{Hom}_R(_RM, {}_RU_S),$$

define

$$f \cdot s : M \to U, \qquad m \mapsto f(m)s.$$

Thus we obtain functors

$$\mathrm{Hom}_R({}_RU_S, -) : R\text{-}\mathbf{Mod} \longrightarrow S\text{-}\mathbf{Mod},$$

given on objects by

$${}_RM \longmapsto \mathrm{Hom}_R({}_RU_S, {}_RM),$$

which is covariant, and

$$\mathrm{Hom}_R(-, {}_RU_S) : R\text{-}\mathbf{Mod} \longrightarrow \mathbf{Mod}\text{-}S,$$

given on objects by

$${}_RM \longmapsto \mathrm{Hom}_R({}_RM, {}_RU_S),$$

which is contravariant.

On morphisms, for

$$f : M \to M',$$

the induced map

$$\mathrm{Hom}_R({}_RU_S, {}_RM) \longrightarrow \mathrm{Hom}_R({}_RU_S, {}_RM')$$

is given by

$$h \longmapsto f \circ h.$$

the induced map

$$\mathrm{Hom}_R({}_RM', {}_RU_S) \longrightarrow \mathrm{Hom}_R({}_RM, {}_RU_S)$$

is given by

$$h \longmapsto h \circ f.$$

Theorem 3.3.2. *The functor*

$$\mathrm{Hom}_R({}_RU_S, -) : R\text{-}\mathbf{Mod} \longrightarrow S\text{-}\mathbf{Mod}$$

is left exact.

Proof. Let

$$0 \longrightarrow N' \xrightarrow{f} N \xrightarrow{g} N''$$

be an exact sequence in R-**Mod**. Consider the induced sequence

$$0 \longrightarrow \mathrm{Hom}_R({}_RU_S, N') \xrightarrow{\mathrm{Hom}_R(U,f)} \mathrm{Hom}_R({}_RU_S, N) \xrightarrow{\mathrm{Hom}_R(U,g)} \mathrm{Hom}_R({}_RU_S, N'').$$

We show that this sequence is exact.

First, $\mathrm{Hom}_R(U, f)$ is injective. Let

$$\varphi \in \mathrm{Hom}_R(U, N')$$

and suppose that

$$\mathrm{Hom}_R(U, f)(\varphi) = f \circ \varphi = 0.$$

Then for every $u \in U$,

$$f(\varphi(u)) = 0.$$

Since f is injective, it follows that $\varphi(u) = 0$ for all $u \in U$. Hence $\varphi = 0$, so $\mathrm{Hom}_R(U, f)$ is injective.

Next, we show that

$$\mathrm{Im}\big(\mathrm{Hom}_R(U, f)\big) \subseteq \ker\big(\mathrm{Hom}_R(U, g)\big).$$

Indeed, if

$$\varphi \in \mathrm{Hom}_R(U, N'),$$

then

$$\mathrm{Hom}_R(U, g)\big(\mathrm{Hom}_R(U, f)(\varphi)\big) = g \circ f \circ \varphi = 0,$$

because $g \circ f = 0$. Therefore

$$\mathrm{Im}\big(\mathrm{Hom}_R(U, f)\big) \subseteq \ker\big(\mathrm{Hom}_R(U, g)\big).$$

Finally, let

$$\psi \in \ker\big(\mathrm{Hom}_R(U, g)\big).$$

Then $\psi \in \mathrm{Hom}_R(U, N)$ and

$$g \circ \psi = 0.$$

Thus for every $u \in U$,

$$g(\psi(u)) = 0,$$

so $\psi(u) \in \ker g = \mathrm{Im}\, f$ by exactness. Hence for each $u \in U$ there exists some $y \in N'$ such that

$$f(y) = \psi(u).$$

Since f is injective, such y is unique. Define

$$\varphi : U \to N', \qquad u \mapsto y,$$

where y is the unique element satisfying $f(y) = \psi(u)$.

Then

$$f \circ \varphi = \psi.$$

It remains to check that φ is R-linear. For $u, v \in U$,

$$f(\varphi(u + v)) = \psi(u + v) = \psi(u) + \psi(v) = f(\varphi(u)) + f(\varphi(v)) = f(\varphi(u) + \varphi(v)).$$

Since f is injective,

$$\varphi(u + v) = \varphi(u) + \varphi(v).$$

Similarly, for $r \in R$ and $u \in U$,

$$f(\varphi(ru)) = \psi(ru) = r\psi(u) = rf(\varphi(u)) = f(r\varphi(u)),$$

so again by injectivity of f,

$$\varphi(ru) = r\varphi(u).$$

Thus $\varphi \in \mathrm{Hom}_R(U, N')$, and

$$\mathrm{Hom}_R(U, f)(\varphi) = f \circ \varphi = \psi.$$

Therefore

$$\ker\bigl(\mathrm{Hom}_R(U, g)\bigr) \subseteq \mathrm{Im}\bigl(\mathrm{Hom}_R(U, f)\bigr).$$

Combining the two inclusions, we get

$$\mathrm{Im}\bigl(\mathrm{Hom}_R(U, f)\bigr) = \ker\bigl(\mathrm{Hom}_R(U, g)\bigr).$$

Hence

$$0 \longrightarrow \mathrm{Hom}_R({}_RU_S, N') \xrightarrow{\mathrm{Hom}_R(U,f)} \mathrm{Hom}_R({}_RU_S, N) \xrightarrow{\mathrm{Hom}_R(U,g)} \mathrm{Hom}_R({}_RU_S, N'')$$

is exact, and so the functor $\mathrm{Hom}_R({}_RU_S, -)$ is left exact. $\square$

Theorem 3.3.3. *Let ${}_RU_S$ be an R-S-bimodule. Then the contravariant functor*

$$\mathrm{Hom}_R(-, U) : R\text{-Mod} \longrightarrow \text{Mod-}S$$

is left exact.

Proof. Since $\mathrm{Hom}_R(-, U)$ is contravariant, to say that it is left exact means the following: for every short exact sequence of left R-modules

$$A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0,$$

the induced sequence of right S-modules

$$0 \longrightarrow \mathrm{Hom}_R(C, U) \xrightarrow{\mathrm{Hom}_R(g,U)} \mathrm{Hom}_R(B, U) \xrightarrow{\mathrm{Hom}_R(f,U)} \mathrm{Hom}_R(A, U)$$

is exact.

We verify this.

First, $\mathrm{Hom}_R(g, U)$ is injective. Indeed, let $\varphi \in \mathrm{Hom}_R(C, U)$ and suppose

$$\mathrm{Hom}_R(g, U)(\varphi) = \varphi \circ g = 0.$$

Since g is surjective, it follows that $\varphi = 0$. Hence $\mathrm{Hom}_R(g, U)$ is injective.

Next, we show
$$\operatorname{Im}\bigl(\operatorname{Hom}_R(g,U)\bigr) \subseteq \ker\bigl(\operatorname{Hom}_R(f,U)\bigr).$$
Let $\varphi \in \operatorname{Hom}_R(C,U)$. Then

$$\operatorname{Hom}_R(f,U)\bigl(\operatorname{Hom}_R(g,U)(\varphi)\bigr) = (\varphi \circ g) \circ f = \varphi \circ (g \circ f) = 0,$$

because $g \circ f = 0$. Therefore

$$\operatorname{Im}\bigl(\operatorname{Hom}_R(g,U)\bigr) \subseteq \ker\bigl(\operatorname{Hom}_R(f,U)\bigr).$$

Finally, we prove the reverse inclusion. Let $\psi \in \operatorname{Hom}_R(B,U)$ satisfy

$$\operatorname{Hom}_R(f,U)(\psi) = \psi \circ f = 0.$$

Then ψ vanishes on $\operatorname{Im}(f) = \ker(g)$. Hence ψ factors through the quotient

$$B/\ker(g).$$

Since g is surjective and $\ker(g) = \operatorname{Im}(f)$, we have

$$B/\ker(g) \cong C.$$

Therefore there exists a unique R-module homomorphism $\overline{\psi} : C \to U$ such that

$$\psi = \overline{\psi} \circ g.$$

Thus
$$\psi \in \operatorname{Im}\bigl(\operatorname{Hom}_R(g,U)\bigr),$$
so
$$\ker\bigl(\operatorname{Hom}_R(f,U)\bigr) \subseteq \operatorname{Im}\bigl(\operatorname{Hom}_R(g,U)\bigr).$$

Combining the two inclusions, we obtain

$$\operatorname{Im}\bigl(\operatorname{Hom}_R(g,U)\bigr) = \ker\bigl(\operatorname{Hom}_R(f,U)\bigr).$$

Hence the sequence

$$0 \longrightarrow \operatorname{Hom}_R(C,U) \xrightarrow{\operatorname{Hom}_R(g,U)} \operatorname{Hom}_R(B,U) \xrightarrow{\operatorname{Hom}_R(f,U)} \operatorname{Hom}_R(A,U)$$

is exact. This proves that $\operatorname{Hom}_R(-,U)$ is left exact. □

Let ${}_SM_R$ be a left-S-right-R-bimodule, ${}_RN$ be a left R-module, consider the tensor product $M \otimes_R N$.

For each $s \in S$, the mapping

$$M \times N \longrightarrow M \otimes_R N, \qquad (m,n) \longmapsto (sm) \otimes n$$

is balanced.

So there is a unique abelian group homomorphism

$$u(s) : M \otimes_R N \longrightarrow M \otimes_R N$$

such that

$$u(s)(m \otimes n) = (sm) \otimes n.$$

Only need to check:

$$u : S \longrightarrow \mathrm{End}_{\mathbb{Z}}(M \otimes_R N)$$

is a ring homomorphism.

If M_R, ${}_RN_T$, then $M \otimes_R N$ is a right T-module:

$$(m \otimes n)t = m \otimes (nt).$$

Lemma 3.3.4. *Let ${}_SM_R$ and ${}_RN_T$ be bimodules. Then $M \otimes_R N$ carries a natural structure of a left S-right T-bimodule, defined on elementary tensors by*

$$s(m \otimes n) = (sm) \otimes n, \qquad (m \otimes n)t = m \otimes (nt),$$

for all $s \in S$, $t \in T$, $m \in M$, and $n \in N$.

Proof. Fix $s \in S$. Define

$$\beta_s : M \times N \to M \otimes_R N, \qquad \beta_s(m, n) = sm \otimes n.$$

We claim that β_s is R-balanced. Indeed, it is additive in each variable, and for $r \in R$,

$$\beta_s(mr, n) = s(mr) \otimes n = (sm)r \otimes n = sm \otimes rn = \beta_s(m, rn).$$

Hence, by the universal property of the tensor product, there exists a unique group homomorphism

$$\lambda_s : M \otimes_R N \to M \otimes_R N$$

such that

$$\lambda_s(m \otimes n) = sm \otimes n.$$

We write $\lambda_s(x) = sx$.

Similarly, for fixed $t \in T$, define

$$\gamma_t : M \times N \to M \otimes_R N, \qquad \gamma_t(m, n) = m \otimes nt.$$

Again γ_t is R-balanced, since for $r \in R$,

$$\gamma_t(mr, n) = mr \otimes nt = m \otimes r(nt) = m \otimes (rn)t = \gamma_t(m, rn).$$

Thus there exists a unique group homomorphism

$$\rho_t : M \otimes_R N \to M \otimes_R N$$

such that

$$\rho_t(m \otimes n) = m \otimes nt.$$

We write $\rho_t(x) = xt$.

It remains to verify the bimodule axioms. Since elementary tensors generate $M \otimes_R N$, it suffices to check them on $m \otimes n$:

$$(s_1 + s_2)(m \otimes n) = ((s_1 + s_2)m) \otimes n = (s_1m) \otimes n + (s_2m) \otimes n = s_1(m \otimes n) + s_2(m \otimes n),$$

$$(s_1s_2)(m \otimes n) = ((s_1s_2)m) \otimes n = s_1((s_2m) \otimes n) = s_1\big(s_2(m \otimes n)\big),$$

$$1_S(m \otimes n) = (1_Sm) \otimes n = m \otimes n,$$

and similarly

$$(m \otimes n)(t_1 + t_2) = m \otimes n(t_1 + t_2) = (m \otimes nt_1) + (m \otimes nt_2),$$

$$((m \otimes n)t_1)t_2 = (m \otimes nt_1)t_2 = m \otimes (nt_1)t_2 = m \otimes n(t_1t_2) = (m \otimes n)(t_1t_2),$$

$$(m \otimes n)1_T = m \otimes n.$$

Finally,

$$\big(s(m \otimes n)\big)t = ((sm) \otimes n)t = (sm) \otimes nt = s(m \otimes nt) = s\big((m \otimes n)t\big).$$

Therefore $M \otimes_R N$ is a left S-right T-bimodule. □

Lemma 3.3.5. *Let M_R, M'_R be right R-modules, and let ${}_RN$, ${}_RN'$ be left R-modules. Suppose*

$$f_1, f_2, f \in \mathrm{Hom}_R(M, M'), \qquad g_1, g_2, g \in \mathrm{Hom}_R(N, N').$$

Then for each pair (f, g) there exists a unique group homomorphism

$$f \otimes g : M \otimes_R N \to M' \otimes_R N'$$

such that

$$(f \otimes g)(m \otimes n) = f(m) \otimes g(n) \qquad (m \in M,\ n \in N).$$

Moreover,

1. *$(f_1 + f_2) \otimes g = (f_1 \otimes g) + (f_2 \otimes g)$;*

2. *$f \otimes (g_1 + g_2) = (f \otimes g_1) + (f \otimes g_2)$;*

3. *$f \otimes 0 = 0$ and $0 \otimes g = 0$;*

4. $1_M \otimes 1_N = 1_{M \otimes_R N}$.

Proof. Define

$$\beta : M \times N \to M' \otimes_R N', \qquad \beta(m,n) = f(m) \otimes g(n).$$

We check that β is R-balanced. It is additive in each variable because f and g are module homomorphisms. Also, for $r \in R$,

$$\beta(mr,n) = f(mr) \otimes g(n) = f(m)r \otimes g(n) = f(m) \otimes rg(n) = f(m) \otimes g(rn) = \beta(m,rn).$$

Hence, by the universal property of $M \otimes_R N$, there exists a unique group homomorphism

$$f \otimes g : M \otimes_R N \to M' \otimes_R N'$$

such that

$$(f \otimes g)(m \otimes n) = f(m) \otimes g(n).$$

Now each asserted identity is verified on elementary tensors, and therefore on all of $M \otimes_R N$.

For (1), for all $m \in M$ and $n \in N$,

$$\big(((f_1 + f_2) \otimes g)(m \otimes n)\big) = (f_1 + f_2)(m) \otimes g(n) = f_1(m) \otimes g(n) + f_2(m) \otimes g(n),$$

while

$$\big((f_1 \otimes g) + (f_2 \otimes g)\big)(m \otimes n) = (f_1 \otimes g)(m \otimes n) + (f_2 \otimes g)(m \otimes n) = f_1(m) \otimes g(n) + f_2(m) \otimes g(n).$$

Hence

$$(f_1 + f_2) \otimes g = (f_1 \otimes g) + (f_2 \otimes g).$$

The proof of (2) is analogous:

$$(f \otimes (g_1 + g_2))(m \otimes n) = f(m) \otimes (g_1 + g_2)(n) = f(m) \otimes g_1(n) + f(m) \otimes g_2(n),$$

which equals

$$((f \otimes g_1) + (f \otimes g_2))(m \otimes n).$$

For (3),

$$(f \otimes 0)(m \otimes n) = f(m) \otimes 0 = 0, \qquad (0 \otimes g)(m \otimes n) = 0 \otimes g(n) = 0,$$

so both maps are the zero homomorphism.

For (4),

$$(1_M \otimes 1_N)(m \otimes n) = 1_M(m) \otimes 1_N(n) = m \otimes n,$$

hence $1_M \otimes 1_N$ is the identity on $M \otimes_R N$. $\square$

Lemma 3.3.6. *Let M_R, M'_R, M''_R be right R-modules, and let ${}_RN$, ${}_RN'$, ${}_RN''$ be left R-modules. Let*

$$f : M \to M', \qquad f' : M' \to M'', \qquad g : N \to N', \qquad g' : N' \to N''$$

be R-module homomorphisms. Then

$$(f' \otimes g') \circ (f \otimes g) = (f' \circ f) \otimes (g' \circ g).$$

Proof. Again it suffices to check the equality on elementary tensors. For $m \in M$ and $n \in N$,

$$\big((f' \otimes g') \circ (f \otimes g)\big)(m \otimes n) = (f' \otimes g')\big(f(m) \otimes g(n)\big) = f'(f(m)) \otimes g'(g(n)),$$

while

$$\big((f' \circ f) \otimes (g' \circ g)\big)(m \otimes n) = (f' \circ f)(m) \otimes (g' \circ g)(n) = f'(f(m)) \otimes g'(g(n)).$$

Thus both homomorphisms agree on the generators $m \otimes n$ of $M \otimes_R N$, hence they are equal. $\square$

Definition 3.3.7. *Let ${}_SU_R$ be a bimodule. Define*

$$U \otimes_R - : R\text{-Mod} \longrightarrow S\text{-Mod}$$

by

$$M \longmapsto U \otimes_R M, \qquad f \longmapsto \mathrm{id}_U \otimes f.$$

Similarly, Let ${}_RU_S$ be a bimodule.

$$- \otimes_R U : \text{Mod-}R \longrightarrow \text{Mod-}S$$

by

$$N \longmapsto N \otimes_R U, \qquad g \longmapsto g \otimes \mathrm{id}_U .$$

Proposition 3.3.8. *Let ${}_SU_R$ be a bimodule. Then*

$$U \otimes_R - : R\text{-Mod} \longrightarrow S\text{-Mod}$$

is a covariant functor.
Similarly, if ${}_RU_S$ is a bimodule, then

$$- \otimes_R U : \text{Mod-}R \longrightarrow \text{Mod-}S$$

is a covariant functor.

Proof. We prove the first statement. For each left R-module M, the tensor product

$$U \otimes_R M$$

is naturally a left S-module via

$$s(u \otimes m) = (su) \otimes m \qquad (s \in S,\ u \in U,\ m \in M).$$

For each R-module homomorphism $f : M \to N$, define

$$U \otimes_R f : U \otimes_R M \to U \otimes_R N$$

by

$$(U \otimes_R f)(u \otimes m) = u \otimes f(m).$$

Equivalently,

$$U \otimes_R f = \mathrm{id}_U \otimes f.$$

This is well-defined and is an S-module homomorphism.

It remains to verify the functorial properties. For the identity map $\mathrm{id}_M : M \to M$, we have

$$U \otimes_R \mathrm{id}_M = \mathrm{id}_U \otimes \mathrm{id}_M = \mathrm{id}_{U \otimes_R M}\,.$$

If

$$M \xrightarrow{f} N \xrightarrow{g} P$$

are R-module homomorphisms, then for every $u \otimes m \in U \otimes_R M$,

$$(U \otimes_R (g \circ f))(u \otimes m) = u \otimes (g \circ f)(m) = g(u \otimes f(m)) = ((U \otimes_R g) \circ (U \otimes_R f))(u \otimes m).$$

Hence

$$U \otimes_R (g \circ f) = (U \otimes_R g) \circ (U \otimes_R f).$$

Therefore $U \otimes_R -$ is a covariant functor from R-Mod to S-Mod.

The proof of the second statement is entirely similar. If ${}_RU_S$ is a bimodule and N is a right R-module, then

$$N \otimes_R U$$

is naturally a right S-module via

$$(n \otimes u)s = n \otimes (us).$$

For a homomorphism $g : N \to N'$ in Mod-R, define

$$g \otimes_R U = g \otimes \mathrm{id}_U : N \otimes_R U \to N' \otimes_R U$$

by

$$(g \otimes_R U)(n \otimes u) = g(n) \otimes u.$$

One checks immediately that identities and compositions are preserved. Thus

$$- \otimes_R U : \text{Mod-}R \to \text{Mod-}S$$

is also a covariant functor. $\square$

Theorem 3.3.9. *Let ${}_SU_R$ be a bimodule. Then the functor*

$$U \otimes_R - \ : \ R\text{-}Mod \longrightarrow S\text{-}Mod$$

is right exact. In other words, if

$$N' \xrightarrow{f} N \xrightarrow{g} N'' \longrightarrow 0$$

is an exact sequence of left R-modules, then

$$U \otimes_R N' \xrightarrow{\mathrm{id}_U \otimes f} U \otimes_R N \xrightarrow{\mathrm{id}_U \otimes g} U \otimes_R N'' \longrightarrow 0$$

is exact.

Proof. Since

$$N' \xrightarrow{f} N \xrightarrow{g} N'' \longrightarrow 0$$

is exact, we have

$$\ker g = \operatorname{Im} f \qquad \text{and} \qquad N'' \cong N/\operatorname{Im} f.$$

Let

$$\pi : N \longrightarrow N/\operatorname{Im} f$$

be the canonical projection. Then there exists an isomorphism

$$\alpha : N/\operatorname{Im} f \longrightarrow N''$$

such that

$$g = \alpha \circ \pi.$$

Now consider the map

$$\Phi : U \otimes_R N \longrightarrow U \otimes_R (N/\operatorname{Im} f), \qquad u \otimes n \longmapsto u \otimes \pi(n).$$

By the universal property of tensor products, Φ is a well-defined S-module homomorphism.

We claim that

$$\ker \Phi = \operatorname{Im}(\mathrm{id}_U \otimes f).$$

First, since $\pi \circ f = 0$, functoriality gives

$$\Phi \circ (\mathrm{id}_U \otimes f) = \mathrm{id}_U \otimes (\pi \circ f) = 0.$$

Hence

$$\operatorname{Im}(\mathrm{id}_U \otimes f) \subseteq \ker \Phi.$$

To prove the reverse inclusion, define

$$\overline{\Phi} : (U \otimes_R N)/\operatorname{Im}(\mathrm{id}_U \otimes f) \longrightarrow U \otimes_R (N/\operatorname{Im} f)$$

by

$$\overline{\Phi}\big((u \otimes n) + \operatorname{Im}(\mathrm{id}_U \otimes f)\big) = u \otimes \pi(n).$$

This is well defined because Φ vanishes on $\operatorname{Im}(\mathrm{id}_U \otimes f)$.

Next define

$$\Psi : U \times (N/\operatorname{Im} f) \longrightarrow (U \otimes_R N)/\operatorname{Im}(\mathrm{id}_U \otimes f)$$

by

$$\Psi(u, \overline{n}) = (u \otimes n) + \operatorname{Im}(\mathrm{id}_U \otimes f),$$

where $\overline{n} = n + \operatorname{Im} f$.

We check that Ψ is well defined. If $\overline{n} = \overline{n'}$, then

$$n - n' \in \operatorname{Im} f,$$

so there exists $x \in N'$ such that

$$n - n' = f(x).$$

Hence

$$(u \otimes n) - (u \otimes n') = u \otimes (n - n') = u \otimes f(x) = (\mathrm{id}_U \otimes f)(u \otimes x) \in \operatorname{Im}(\mathrm{id}_U \otimes f).$$

Thus

$$(u \otimes n) + \operatorname{Im}(\mathrm{id}_U \otimes f) = (u \otimes n') + \operatorname{Im}(\mathrm{id}_U \otimes f).$$

So Ψ is well defined. It is clearly R-balanced, hence induces an S-module homomorphism

$$\overline{\Psi} : U \otimes_R (N/\operatorname{Im} f) \longrightarrow (U \otimes_R N)/\operatorname{Im}(\mathrm{id}_U \otimes f)$$

such that

$$\overline{\Psi}(u \otimes \overline{n}) = (u \otimes n) + \operatorname{Im}(\mathrm{id}_U \otimes f).$$

It is immediate that $\overline{\Phi}$ and $\overline{\Psi}$ are inverse to each other. Therefore

$$(U \otimes_R N)/\operatorname{Im}(\mathrm{id}_U \otimes f) \cong U \otimes_R (N/\operatorname{Im} f).$$

It follows that

$$\ker \Phi = \operatorname{Im}(\mathrm{id}_U \otimes f).$$

Finally, since $\alpha : N/\operatorname{Im} f \to N''$ is an isomorphism, the induced map

$$\mathrm{id}_U \otimes \alpha : U \otimes_R (N/\operatorname{Im} f) \longrightarrow U \otimes_R N''$$

is an isomorphism. Moreover,

$$\mathrm{id}_U \otimes g = (\mathrm{id}_U \otimes \alpha) \circ \Phi.$$

Hence

$$\ker(\mathrm{id}_U \otimes g) = \ker \Phi = \mathrm{Im}(\mathrm{id}_U \otimes f),$$

and $\mathrm{id}_U \otimes g$ is surjective because Φ is surjective and $\mathrm{id}_U \otimes \alpha$ is an isomorphism.

Therefore

$$U \otimes_R N' \xrightarrow{\mathrm{id}_U \otimes f} U \otimes_R N \xrightarrow{\mathrm{id}_U \otimes g} U \otimes_R N'' \longrightarrow 0$$

is exact. $\square$

Chapter 4

Homework

4.1 Homework Week 1

Exercise 4.1.1. *Let R be a ring, and let M be an R-module. Suppose that S is a ring and*

$$\theta : S \to R$$

is a ring homomorphism satisfying $\theta(1_S) = 1_R$. Let S act on M by

$$sx = \theta(s)x,$$

where $x \in M$ and $s \in S$. Prove that M is an S-module.

Proof. This action is well-defined because $\forall s \in S$, $\theta(s) \in R$ and $\theta(s)x \in M$ for all $x \in M$. We need to verify the module axioms, for all $s, t \in S$ and $x, y \in M$:

- $s(x + y) = \theta(s)(x + y) = \theta(s)x + \theta(s)y = sx + sy$
- $(s + t)x = \theta(s + t)x = \theta(s)x + \theta(t)x = sx + tx$
- $(st)x = \theta(st)x = \theta(s)\theta(t)x = \theta(s)(\theta(t)x) = s(tx)$.
- $1_S x = \theta(1_S)x = 1_R x = x$

Hence M is an S-module. □

Exercise 4.1.2. *Let R be a ring, and let M and N be two right R-modules. Let $\mathrm{Hom}_{\mathbb{Z}}(M, N)$ be the set of group homomorphisms from M to N (as abelian groups). The action of R on $\mathrm{Hom}_{\mathbb{Z}}(M, N)$ is given as follows: for any $a \in R$ and $\varphi \in \mathrm{Hom}_{\mathbb{Z}}(M, N)$, define*

$$(\varphi a)(x) = \varphi(xa), \qquad \forall x \in M.$$

Prove that $\mathrm{Hom}_{\mathbb{Z}}(M, N)$ is a left R-module.

Proof. This action is well-defined. $\forall r \in R$, $\varphi r \in \mathrm{Hom}_{\mathbb{Z}}(M, N)$ for all $\varphi \in \mathrm{Hom}_{\mathbb{Z}}(M, N)$ as $(\varphi r)(x) = \varphi(xr) \in N$ for all $x \in M$. We need to verify the module axioms, for all $r_1, r_2 \in R$ and $\varphi, \psi \in \mathrm{Hom}_{\mathbb{Z}}(M, N)$, $\forall x \in M$:

- $((\varphi + \psi)r)(x) = (\varphi + \psi)(xr) = \varphi(xr) + \psi(xr) = (\varphi r)(x) + (\psi r)(x)$
- $(\varphi(r_1+r_2))(x) = \varphi(x(r_1+r_2)) = \varphi(xr_1+xr_2) = \varphi(xr_1)+\varphi(xr_2) = (\varphi r_1)(x)+(\varphi r_2)(x)$
- $((\varphi r_1)r_2)(x) = (\varphi r_2)(xr_1) = \varphi(xr_2r_1) = (\varphi r_2 r_1)(x)$
- $(\varphi 1_R)(x) = \varphi(x1_R) = \varphi(x)$

Hence $\operatorname{Hom}_{\mathbb{Z}}(M, N)$ is a left R-module. □

Exercise 4.1.3. *Let R be a ring, and let M be an R-module. Denote*

$$\operatorname{Ann}_R M = \{r \in R \mid rm = 0,\ \forall m \in M\}.$$

We call M a faithful R-module if $\operatorname{Ann}_R M = \{0\}$. *Define the action of the quotient ring $R/\operatorname{Ann}_R M$ on M by*

$$\overline{r}\,x = rx,$$

where $x \in M$, $r \in R$, and $\overline{r} = r + \operatorname{Ann}_R M$. Prove that M is a faithful $R/\operatorname{Ann}_R M$-module.

Proof. First we show that the action is well-defined. Suppose

$$\bar{r} = \bar{r}' \in R/\operatorname{Ann}_R M.$$

Then $r - r' \in \operatorname{Ann}_R M$, so for every $x \in M$,

$$(r - r')x = 0.$$

Hence $rx = r'x$, so $\bar{r}\,x$ does not depend on the choice of representative.

The module axioms follow immediately from those of the R-module structure on M.

Now we prove faithfulness. Suppose $\bar{r} \in R/\operatorname{Ann}_R M$ acts trivially on M, that is,

$$\bar{r}\,x = 0 \qquad \forall x \in M.$$

By definition, this means

$$rx = 0 \qquad \forall x \in M.$$

Therefore $r \in \operatorname{Ann}_R M$, so $\bar{r} = \bar{0}$ in the quotient ring. Hence

$$\operatorname{Ann}_{R/\operatorname{Ann}_R M}(M) = \{\bar{0}\},$$

and M is faithful as an $R/\operatorname{Ann}_R M$-module. □

Exercise 4.1.4. *Let $\varphi : M \to M$ be a homomorphism of R-modules such that*

$$\varphi\varphi = \varphi.$$

Prove that

$$M = \ker\varphi \oplus \operatorname{im}\varphi.$$

Proof. Let $x \in M$. Then $x = \varphi(x) + (x - \varphi(x))$. Where $\varphi(x) \in \operatorname{im}\varphi$ and $x - \varphi(x) \in \ker\varphi$. Hence $M = \operatorname{im}\varphi + \ker\varphi$. We only need to show that $\ker\varphi \cap \operatorname{im}\varphi = \{0\}$. Suppose $x \in \ker\varphi \cap \operatorname{im}\varphi$. Then $x = \varphi(y)$ for some $y \in M$ and $\varphi(x) = 0$. Then $\varphi(x) = \varphi\varphi(y) = \varphi(y) = x = 0$, so $x = 0$. Hence $\ker\varphi \cap \operatorname{im}\varphi = \{0\}$. □

Exercise 4.1.5. *Show that any R-module is a homomorphism image of some free R-module.*

Proof. Let M be a left R-module. Consider the free R-module

$$F = \bigoplus_{x \in M} Re_x$$

with basis $\{e_x\}_{x \in M}$ indexed by the underlying set of M. Define an R-module homomorphism

$$\pi : F \to M$$

by

$$\pi(e_x) = x \qquad (x \in M),$$

and extend R-linearly. Thus

$$\pi\Big(\sum_{i=1}^{n} r_i e_{x_i}\Big) = \sum_{i=1}^{n} r_i x_i.$$

This is well-defined and R-linear.

For any $m \in M$, we have

$$\pi(e_m) = m,$$

so π is surjective. Therefore M is a homomorphic image of the free module F. Equivalently,

$$M \cong F/\ker\pi.$$

□

Exercise 4.1.6. *Let $\mathbb{Q}$ be the field of rational numbers. Is the additive group $(\mathbb{Q}, +)$ a free $\mathbb{Z}$-module?*

Proof. Claim: any two non-zero element of $\mathbb{Q}$ are linearly dependent over $\mathbb{Z}$. Hence if $\mathbb{Q}$ is a free $\mathbb{Z}$-module, then it must have a basis of cardinality 1.

Proof of claim: let $0 \neq x = \frac{a}{b}$ and $0 \neq y = \frac{c}{d}$ be two elements of $\mathbb{Q}$. Then $dcx - aby = 0$. If $a = 0$ or $c = 0$, then $x = 0$ or $y = 0$, which contradicts the assumption. Hence $a, c \neq 0$ which means dc, ab are two non-zero integer coefficients. Hence x and y are linearly dependent over $\mathbb{Z}$. And if $\mathbb{Q}$ is a free $\mathbb{Z}$-module, then it must have a basis of cardinality 1.

But if $\mathbb{Q}$ is a cyclic $\mathbb{Z}$-module, then it has only one generator. Suppose it is $z = \frac{e}{f}$. Then $\mathbb{Q} = \{n\frac{e}{f} \mid n \in \mathbb{Z}\}$. But this is not possible because we can easily pick $\frac{1}{2f} \notin \mathbb{Q}$, which is absurd.

Therefore $\mathbb{Q}$ is not a free $\mathbb{Z}$-module. □

4.2 Homework Week 2

Exercise 4.2.1. *Let R be a commutative ring, and suppose that φ is an endomorphism of the free R-module R^m. Prove that if φ is surjective, then φ is injective. Consider that: if φ is injective, is φ surjective?*

Proof. Let $\{e_1, \ldots, e_m\}$ be the standard basis of R^m, and let $A \in M_m(R)$ be the matrix of φ with respect to this basis. Since φ is surjective, for each j there exists $v_j \in R^m$ with $\varphi(v_j) = e_j$. Write $v_j = \sum_{i=1}^m b_{ij} e_i$ and let $B = (b_{ij}) \in M_m(R)$. Then $AB = I_m$, so

$$\det(A)\det(B) = \det(AB) = 1.$$

Hence $\det(A)$ is a unit in R. It follows that A is invertible (with $A^{-1} = \det(A)^{-1} \operatorname{adj}(A)$), so φ is an isomorphism. In particular φ is injective.

Counterexample for the converse. The converse is false in general. Let $R = \mathbb{Z}$ and $m = 1$, so $R^m = \mathbb{Z}$. Define $\varphi : \mathbb{Z} \to \mathbb{Z}$ by $\varphi(n) = 2n$. Then φ is injective but not surjective (e.g. $1 \notin \operatorname{im} \varphi$). □

Exercise 4.2.2. *Let $\varphi : M \to N$ be a homomorphism of R-modules. Prove that:*

1. *φ is a monomorphism if and only if for any R-module T and any two R-module homomorphisms $\xi, \eta : T \to M$, $\varphi\xi = \varphi\eta$ implies $\xi = \eta$;*

2. *φ is an epimorphism if and only if for any R-module S and any two R-module homomorphisms $\rho, \theta : N \to S$, $\rho\varphi = \theta\varphi$ implies $\rho = \theta$.*

Proof.

1. ($\Rightarrow$) Suppose φ is injective and $\varphi\xi = \varphi\eta$ for $\xi, \eta : T \to M$. Then for every $t \in T$, $\varphi(\xi(t)) = \varphi(\eta(t))$, so $\xi(t) - \eta(t) \in \ker \varphi = \{0\}$. Hence $\xi(t) = \eta(t)$ for all t, i.e. $\xi = \eta$.

   ($\Leftarrow$) We show $\ker \varphi = \{0\}$. Let $T = \ker \varphi$, $\xi : T \hookrightarrow M$ be the inclusion, and $\eta = 0 : T \to M$ the zero map. Then $\varphi\xi(t) = \varphi(t) = 0 = \varphi\eta(t)$ for all $t \in T$, so $\varphi\xi = \varphi\eta$. By hypothesis $\xi = \eta$, which means $t = \xi(t) = \eta(t) = 0$ for all $t \in T$. Hence $\ker \varphi = \{0\}$, so φ is injective.

2. ($\Rightarrow$) Suppose φ is surjective and $\rho\varphi = \theta\varphi$. For any $n \in N$, pick $m \in M$ with $\varphi(m) = n$. Then $\rho(n) = \rho(\varphi(m)) = \theta(\varphi(m)) = \theta(n)$. Hence $\rho = \theta$.

   ($\Leftarrow$) We show φ is surjective. Let $S = N/\operatorname{im} \varphi$, $\rho : N \to S$ the canonical projection, and $\theta = 0 : N \to S$. For any $m \in M$, $\rho(\varphi(m)) = \overline{\varphi(m)} = \bar{0} = \theta(\varphi(m))$, so $\rho\varphi = \theta\varphi$. By hypothesis $\rho = \theta$, i.e. the projection $N \to N/\operatorname{im} \varphi$ is the zero map. This means $N = \operatorname{im} \varphi$, so φ is surjective.

□

Exercise 4.2.3. *Let M and M_i $(1 \leq i \leq n)$ be R-modules. Show that $M \simeq \bigoplus_{i=1}^n M_i$ if and only if for every $i \in \{1, \ldots, n\}$, there exists an $\varphi_i \in \operatorname{End}_R(M)$ such that $\operatorname{im} \varphi_i \simeq M_i$, $\varphi_i\varphi_j = 0$ for $i \neq j$, $\varphi_i^2 = \varphi_i$, and $\varphi_1 + \varphi_2 + \cdots + \varphi_n = \operatorname{id}_M$.*

Proof. ($\Rightarrow$) Suppose $f : M \xrightarrow{\sim} \bigoplus_{i=1}^n M_i$. Let $\iota_i : M_i \hookrightarrow \bigoplus_{j=1}^n M_j$ be the canonical inclusion and $\pi_i : \bigoplus_{j=1}^n M_j \twoheadrightarrow M_i$ the canonical projection. Define

$$\varphi_i = f^{-1} \circ \iota_i \circ \pi_i \circ f \in \mathrm{End}_R(M).$$

Then:

- $\varphi_i^2 = f^{-1}\iota_i\pi_i f f^{-1}\iota_i\pi_i f = f^{-1}\iota_i(\pi_i\iota_i)\pi_i f = f^{-1}\iota_i\pi_i f = \varphi_i$, since $\pi_i\iota_i = \mathrm{id}_{M_i}$.
- For $i \neq j$: $\varphi_i\varphi_j = f^{-1}\iota_i(\pi_i\iota_j)\pi_j f = 0$, since $\pi_i\iota_j = 0$ when $i \neq j$.
- $\sum_{i=1}^n \varphi_i = f^{-1}\left(\sum_{i=1}^n \iota_i\pi_i\right) f = f^{-1} \circ \mathrm{id} \circ f = \mathrm{id}_M$, since $\sum_i \iota_i\pi_i = \mathrm{id}_{\bigoplus M_j}$.
- $\operatorname{im}\varphi_i = f^{-1} \circ \iota_i \circ \pi_i \circ f(M) \simeq M_i$.

($\Leftarrow$) Suppose such φ_i exist. Define $g : M \to \bigoplus_{i=1}^n \operatorname{im}\varphi_i$ by $g(x) = (\varphi_1(x), \ldots, \varphi_n(x))$. *Surjective*: for any $(y_1, \ldots, y_n) \in \bigoplus \operatorname{im}\varphi_i$, let $x = y_1 + \cdots + y_n \in M$. Then

$$\varphi_j(x) = \sum_{i=1}^n \varphi_j(y_i) = \sum_{i=1}^n \varphi_j(\varphi_i(x_i)) = \varphi_j(\varphi_j(x_j)) = \varphi_j^2(x_j) = \varphi_j(x_j) = y_j,$$

where we used $\varphi_j\varphi_i = 0$ for $j \neq i$ and $\varphi_j^2 = \varphi_j$. Hence $g(x) = (y_1, \ldots, y_n)$.

Injective: if $g(x) = 0$, then $\varphi_i(x) = 0$ for all i, so $x = \mathrm{id}_M(x) = \sum_i \varphi_i(x) = 0$.

Since $\operatorname{im}\varphi_i \simeq M_i$, we conclude $M \simeq \bigoplus_{i=1}^n M_i$. □

Exercise 4.2.4.

1. *Prove $\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \simeq \mathbb{Z}/(m,n)\mathbb{Z}$, where $m, n \in \mathbb{Z}$.*
2. *Let A be an abelian group, prove $A \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \simeq A/nA$.*
3. *Let A, B be two finitely generated abelian groups. What is $A \otimes_{\mathbb{Z}} B$?*

Proof.

1. We use the right-exact sequence $\mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z} \to 0$. Tensoring with $\mathbb{Z}/m\mathbb{Z}$ over $\mathbb{Z}$ gives the right-exact sequence

$$\mathbb{Z}/m\mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z}/m\mathbb{Z} \to \mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \to 0.$$

Hence $\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \simeq (\mathbb{Z}/m\mathbb{Z})/n(\mathbb{Z}/m\mathbb{Z})$. Now $n(\mathbb{Z}/m\mathbb{Z}) = (n\mathbb{Z}+m\mathbb{Z})/m\mathbb{Z}$. Since $n\mathbb{Z} + m\mathbb{Z} = (m,n)\mathbb{Z}$, we get

$$\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \simeq \frac{\mathbb{Z}/m\mathbb{Z}}{(m,n)\mathbb{Z}/m\mathbb{Z}} \simeq \mathbb{Z}/(m,n)\mathbb{Z}.$$

2. Similarly, tensor the right-exact sequence $\mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z} \to 0$ with A:

$$A \xrightarrow{\cdot n} A \to A \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \to 0.$$

By right-exactness, $A \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \simeq A/nA$.

3. By the fundamental theorem of finitely generated abelian groups,

$$A \simeq \mathbb{Z}^r \oplus \bigoplus_{i=1}^{s} \mathbb{Z}/d_i\mathbb{Z}, \qquad B \simeq \mathbb{Z}^t \oplus \bigoplus_{j=1}^{u} \mathbb{Z}/e_j\mathbb{Z}.$$

Since tensor product distributes over direct sums, using (1), (2), and $\mathbb{Z} \otimes_{\mathbb{Z}} M \simeq M$, we get

$$A \otimes_{\mathbb{Z}} B \simeq \mathbb{Z}^{rt} \oplus \bigoplus_{j=1}^{u}(\mathbb{Z}/e_j\mathbb{Z})^r \oplus \bigoplus_{i=1}^{s}(\mathbb{Z}/d_i\mathbb{Z})^t \oplus \bigoplus_{i=1}^{s}\bigoplus_{j=1}^{u} \mathbb{Z}/(d_i, e_j)\mathbb{Z}.$$

$\square$

Exercise 4.2.5. *Let $\varphi : \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/4\mathbb{Z}$ be the unique monomorphism. Prove that*

$$\mathrm{id} \otimes \varphi : \mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/4\mathbb{Z}$$

is a zero homomorphism.

Proof. The unique monomorphism $\varphi : \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/4\mathbb{Z}$ is given by $\varphi(\bar{1}) = \bar{2}$. By the previous exercise, $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \simeq \mathbb{Z}/2\mathbb{Z}$ and $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/4\mathbb{Z} \simeq \mathbb{Z}/2\mathbb{Z}$.

It suffices to check on the generator $\bar{1} \otimes \bar{1}$ of $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z}$:

$$(\mathrm{id} \otimes \varphi)(\bar{1} \otimes \bar{1}) = \bar{1} \otimes \varphi(\bar{1}) = \bar{1} \otimes \bar{2} = \bar{1} \otimes 2 \cdot \bar{1} = 2 \cdot (\bar{1} \otimes \bar{1}) = (2 \cdot \bar{1}) \otimes \bar{1} = \bar{0} \otimes \bar{1} = 0,$$

where we used $2 \cdot \bar{1} = \bar{0}$ in $\mathbb{Z}/2\mathbb{Z}$. Since the generator maps to 0, the map $\mathrm{id} \otimes \varphi$ is the zero homomorphism.

This also shows that tensoring a monomorphism need not yield a monomorphism, i.e. the functor $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} -$ is not left-exact. $\square$

Exercise 4.2.6. *Let V_1 and V_2 be two vector spaces over a field F with dimension n and m respectively. Let $\{\epsilon_1, \dots, \epsilon_n\}$ and $\{\eta_1, \dots, \eta_m\}$ be bases of V_1 and V_2 respectively. Suppose that $\mathscr{A}$ (resp. $\mathscr{B}$) is an F-linear map of V_1 (resp. V_2), and it is represented by the matrix $A = (a_{ij})_{n\times n}$ (resp. $B = (b_{ij})_{m\times m}$) with respect to the above bases. Prove that*

$$\{\epsilon_i \otimes \eta_j \mid 1 \le i \le n,\ 1 \le j \le m\}$$

is an F-basis of $V_1 \otimes_F V_2$. Determine the matrix representing the F-linear map $\mathscr{A} \otimes \mathscr{B}$ under this basis.

Proof. **Step 1: $\{\epsilon_i \otimes \eta_j\}$ is a basis.**

Spanning. Every element of $V_1 \otimes_F V_2$ is a finite sum of simple tensors $v \otimes w$. Writing $v = \sum_i a_i\epsilon_i$ and $w = \sum_j b_j\eta_j$, by bilinearity of the tensor product we get

$$v \otimes w = \sum_{i,j} a_i b_j\, (\epsilon_i \otimes \eta_j).$$

Hence $\{\epsilon_i \otimes \eta_j\}$ spans $V_1 \otimes_F V_2$.

Linear independence. Suppose $\sum_{i,j} c_{ij}(\epsilon_i \otimes \eta_j) = 0$. For each pair (p, q), let $f_p \in V_1^*$ and $g_q \in V_2^*$ be the dual basis functionals: $f_p(\epsilon_i) = \delta_{pi}$ and $g_q(\eta_j) = \delta_{qj}$. The bilinear map $(v, w) \mapsto f_p(v)g_q(w)$ from $V_1 \times V_2$ to F induces (by the universal property of the tensor product) an F-linear map $\Phi_{pq} : V_1 \otimes_F V_2 \to F$ with $\Phi_{pq}(\epsilon_i \otimes \eta_j) = \delta_{pi}\delta_{qj}$. Applying Φ_{pq} to the relation $\sum_{i,j} c_{ij}(\epsilon_i \otimes \eta_j) = 0$ yields $c_{pq} = 0$. Since (p, q) was arbitrary, all $c_{ij} = 0$.

Therefore $\{\epsilon_i \otimes \eta_j : 1 \le i \le n,\ 1 \le j \le m\}$ is an F-basis of $V_1 \otimes_F V_2$, and $\dim_F(V_1 \otimes_F V_2) = nm$.

Step 2: matrix of $\mathscr{A} \otimes \mathscr{B}$.

Order the basis lexicographically: $\epsilon_1 \otimes \eta_1,\ \epsilon_1 \otimes \eta_2,\ \ldots,\ \epsilon_1 \otimes \eta_m,\ \epsilon_2 \otimes \eta_1,\ \ldots,\ \epsilon_n \otimes \eta_m$. Since $\mathscr{A}(\epsilon_i) = \sum_{k=1}^n a_{ki}\epsilon_k$ and $\mathscr{B}(\eta_j) = \sum_{l=1}^m b_{lj}\eta_l$, we have

$$(\mathscr{A} \otimes \mathscr{B})(\epsilon_i \otimes \eta_j) = \mathscr{A}(\epsilon_i) \otimes \mathscr{B}(\eta_j) = \sum_{k=1}^n \sum_{l=1}^m a_{ki}b_{lj}\,(\epsilon_k \otimes \eta_l).$$

The entry in row (k, l) and column (i, j) of the representing matrix is $a_{ki}b_{lj}$. Written as an $nm \times nm$ block matrix, this is the **Kronecker product**

$$A \otimes B = \begin{pmatrix} a_{11}B & a_{12}B & \cdots & a_{1n}B \\ a_{21}B & a_{22}B & \cdots & a_{2n}B \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1}B & a_{n2}B & \cdots & a_{nn}B \end{pmatrix}.$$

□

4.3 Homework Week 3

Exercise 4.3.1. *Let M, N, M_i ($i \in I$) and N_j ($j \in J$) be R-modules. Prove the following isomorphisms of abelian groups:*

1.

$$\operatorname{Hom}_R\left(\bigoplus_{i\in I} M_i,\ N\right) \cong \prod_{i\in I} \operatorname{Hom}_R(M_i, N).$$

2.

$$\operatorname{Hom}_R\left(M,\ \prod_{j\in J} N_j\right) \cong \prod_{j\in J} \operatorname{Hom}_R(M, N_j).$$

Proof. We prove the two statements separately.

(1) Define

$$\Phi : \operatorname{Hom}_R\left(\bigoplus_{i\in I} M_i,\ N\right) \longrightarrow \prod_{i\in I} \operatorname{Hom}_R(M_i, N)$$

by

$$\Phi(f) = (f \circ \iota_i)_{i\in I},$$

where

$$\iota_i : M_i \to \bigoplus_{i\in I} M_i$$

is the canonical inclusion.

We show that Φ is an isomorphism.

First, Φ is a group homomorphism, since composition is compatible with addition:

$$\Phi(f+g) = \big((f+g)\circ\iota_i\big)_{i\in I} = \big(f\circ\iota_i + g\circ\iota_i\big)_{i\in I} = \Phi(f)+\Phi(g).$$

Now let $(f_i)_{i\in I} \in \prod_{i\in I} \mathrm{Hom}_R(M_i, N)$. Define

$$f : \bigoplus_{i\in I} M_i \to N$$

by

$$f\big((m_i)_{i\in I}\big) = \sum_{i\in I} f_i(m_i).$$

This is well-defined, because an element of $\bigoplus_{i\in I} M_i$ has only finitely many nonzero components, so the above sum is finite. It is immediate that f is R-linear, and

$$f\circ\iota_i = f_i \qquad \text{for all } i\in I.$$

Hence $\Phi(f) = (f_i)_{i\in I}$, so Φ is surjective.

To prove injectivity, suppose $\Phi(f)=\Phi(g)$. Then

$$f\circ\iota_i = g\circ\iota_i \qquad \text{for all } i\in I.$$

Take any $x = (m_i)_{i\in I} \in \bigoplus_{i\in I} M_i$. Since x has finite support, we may write

$$x = \sum_{i\in I} \iota_i(m_i),$$

with only finitely many nonzero terms. Therefore,

$$f(x) = \sum_{i\in I} f(\iota_i(m_i)) = \sum_{i\in I} g(\iota_i(m_i)) = g(x).$$

So $f=g$, and Φ is injective.

Thus Φ is an isomorphism:

$$\mathrm{Hom}_R\left(\bigoplus_{i\in I} M_i,\, N\right) \cong \prod_{i\in I} \mathrm{Hom}_R(M_i, N).$$

(2) Let

$$\pi_j : \prod_{j\in J} N_j \to N_j$$

be the canonical projection. Define

$$\Psi : \mathrm{Hom}_R\left(M, \prod_{j\in J} N_j\right) \longrightarrow \prod_{j\in J} \mathrm{Hom}_R(M, N_j)$$

by

$$\Psi(f) = (\pi_j \circ f)_{j\in J}.$$

Again, Ψ is a group homomorphism.

We show that Ψ is an isomorphism. Given a family

$$(g_j)_{j\in J} \in \prod_{j\in J} \mathrm{Hom}_R(M, N_j),$$

define

$$g : M \to \prod_{j\in J} N_j$$

by

$$g(m) = (g_j(m))_{j\in J}.$$

Since each g_j is R-linear, g is also R-linear. Moreover,

$$\pi_j \circ g = g_j \qquad \text{for all } j \in J,$$

so

$$\Psi(g) = (g_j)_{j\in J}.$$

Hence Ψ is surjective.

For injectivity, suppose $\Psi(f) = \Psi(g)$. Then

$$\pi_j \circ f = \pi_j \circ g \qquad \text{for all } j \in J.$$

Thus for every $m \in M$ and every $j \in J$,

$$\pi_j(f(m)) = \pi_j(g(m)).$$

Since two elements of $\prod_{j\in J} N_j$ are equal if and only if all their components are equal, it follows that

$$f(m) = g(m) \qquad \text{for all } m \in M.$$

Hence $f = g$, and Ψ is injective.

Therefore

$$\mathrm{Hom}_R\left(M, \prod_{j\in J} N_j\right) \cong \prod_{j\in J} \mathrm{Hom}_R(M, N_j).$$

This completes the proof. $\square$

Exercise 4.3.2 (Five Lemma)**.** *Suppose that the following diagram of homomorphisms of*

R-modules

$$\begin{tikzcd}
M_1 \arrow[r] \arrow[d, "\varphi_1"'] & M_2 \arrow[r] \arrow[d, "\varphi_2"'] & M_3 \arrow[r] \arrow[d, "\varphi_3"'] & M_4 \arrow[r] \arrow[d, "\varphi_4"'] & M_5 \arrow[d, "\varphi_5"'] \\
N_1 \arrow[r] & N_2 \arrow[r] & N_3 \arrow[r] & N_4 \arrow[r] & N_5
\end{tikzcd}$$

is commutative and has exact rows. Prove that:

1. *If φ_1 is surjective, and φ_2 and φ_4 are injective, then φ_3 is injective.*
2. *If φ_5 is injective, and φ_2 and φ_4 are surjective, then φ_3 is surjective.*

Proof. We prove the two assertions separately.

(1) Assume that φ_1 is surjective and that φ_2, φ_4 are injective. We show that φ_3 is injective.

Let $x \in M_3$ and suppose that

$$\varphi_3(x) = 0.$$

We must prove that $x = 0$.

Let $\alpha_3 : M_3 \to M_4$ and $\beta_3 : N_3 \to N_4$ denote the third horizontal maps. By commutativity,

$$\varphi_4(\alpha_3(x)) = \beta_3(\varphi_3(x)) = \beta_3(0) = 0.$$

Since φ_4 is injective, it follows that $\alpha_3(x) = 0$. Exactness of the top row at M_3 yields

$$x = \alpha_2(y)$$

for some $y \in M_2$, where $\alpha_2 : M_2 \to M_3$ is the second horizontal map.

Again by commutativity,

$$0 = \varphi_3(x) = \varphi_3(\alpha_2(y)) = \beta_2(\varphi_2(y)),$$

where $\beta_2 : N_2 \to N_3$ is the second horizontal map. Hence $\varphi_2(y) \in \ker(\beta_2)$. By exactness of the bottom row at N_2,

$$\ker(\beta_2) = \operatorname{im}(\beta_1),$$

so there exists $z \in N_1$ such that

$$\beta_1(z) = \varphi_2(y),$$

where $\beta_1 : N_1 \to N_2$ is the first horizontal map.

Since φ_1 is surjective, there exists $w \in M_1$ such that

$$\varphi_1(w) = z.$$

Using commutativity once more,

$$\varphi_2(\alpha_1(w)) = \beta_1(\varphi_1(w)) = \beta_1(z) = \varphi_2(y),$$

where $\alpha_1 : M_1 \to M_2$ is the first horizontal map. Because φ_2 is injective, we get

$$\alpha_1(w) = y.$$

Therefore

$$x = \alpha_2(y) = \alpha_2(\alpha_1(w)) = 0,$$

since the composition of two consecutive maps in an exact sequence is zero. Thus $\ker(\varphi_3) = 0$, so φ_3 is injective.

(2) Assume that φ_5 is injective and that φ_2, φ_4 are surjective. We show that φ_3 is surjective.

Let $n \in N_3$. We must find $m \in M_3$ such that

$$\varphi_3(m) = n.$$

Let $\beta_3 : N_3 \to N_4$ and $\alpha_3 : M_3 \to M_4$ denote the third horizontal maps. Since φ_4 is surjective, there exists $u \in M_4$ such that

$$\varphi_4(u) = \beta_3(n).$$

Let $\alpha_4 : M_4 \to M_5$ and $\beta_4 : N_4 \to N_5$ be the next horizontal maps. Then

$$\varphi_5(\alpha_4(u)) = \beta_4(\varphi_4(u)) = \beta_4(\beta_3(n)) = 0.$$

Because φ_5 is injective, it follows that $\alpha_4(u) = 0$. Exactness of the top row at M_4 shows that there exists $m \in M_3$ such that

$$\alpha_3(m) = u.$$

Consider

$$n - \varphi_3(m) \in N_3.$$

By commutativity,

$$\beta_3\big(n - \varphi_3(m)\big) = \beta_3(n) - \beta_3(\varphi_3(m)) = \beta_3(n) - \varphi_4(\alpha_3(m)) = \beta_3(n) - \varphi_4(u) = 0.$$

Hence $n - \varphi_3(m) \in \ker(\beta_3)$. By exactness of the bottom row at N_3,

$$\ker(\beta_3) = \operatorname{im}(\beta_2),$$

so there exists $v \in N_2$ such that

$$\beta_2(v) = n - \varphi_3(m),$$

where $\beta_2 : N_2 \to N_3$ is the second horizontal map.

Since φ_2 is surjective, there exists $y \in M_2$ such that

$$\varphi_2(y) = v.$$

Then

$$\varphi_3(\alpha_2(y)) = \beta_2(\varphi_2(y)) = \beta_2(v) = n - \varphi_3(m),$$

where $\alpha_2 : M_2 \to M_3$ is the second horizontal map. Therefore

$$n = \varphi_3(m) + \varphi_3(\alpha_2(y)) = \varphi_3\big(m + \alpha_2(y)\big).$$

Thus n lies in the image of φ_3, proving that φ_3 is surjective. $\square$

Exercise 4.3.3 (Snake Lemma)**.** *Suppose that the following diagram of homomorphisms of R-modules*

$$\begin{tikzcd}
0 \arrow[r] & M' \arrow[r, "\alpha"] \arrow[d, "\varphi'"'] & M \arrow[r, "\beta"] \arrow[d, "\varphi"'] & M'' \arrow[r] \arrow[d, "\varphi''"'] & 0 \\
0 \arrow[r] & N' \arrow[r, "\mu"] & N \arrow[r, "\nu"] & N'' \arrow[r] & 0
\end{tikzcd}$$

is commutative and has exact rows. Prove that

$$0 \longrightarrow \ker\varphi' \xrightarrow{\alpha'} \ker\varphi \xrightarrow{\beta'} \ker\varphi'' \xrightarrow{\delta} \operatorname{coker}\varphi' \xrightarrow{\mu'} \operatorname{coker}\varphi \xrightarrow{\nu'} \operatorname{coker}\varphi'' \longrightarrow 0$$

is a long exact sequence, where $\alpha', \beta', \mu', \nu'$ are the homomorphisms induced by α, β, μ, ν, respectively.

Proof. Since the rows are exact, α and μ are injective, while β and ν are surjective.

We first define the connecting homomorphism

$$\delta : \ker\varphi'' \to \operatorname{coker}\varphi'.$$

Let $x \in \ker\varphi'' \subseteq M''$. Since $\beta : M \to M''$ is surjective, choose $y \in M$ such that

$$\beta(y) = x.$$

By commutativity,

$$\nu(\varphi(y)) = \varphi''(\beta(y)) = \varphi''(x) = 0.$$

Hence $\varphi(y) \in \ker\nu = \operatorname{im}\mu$. Therefore there exists $z \in N'$ such that

$$\mu(z) = \varphi(y).$$

Define

$$\delta(x) = \overline{z} = z + \operatorname{im}\varphi' \in \operatorname{coker}\varphi'.$$

We must check that this is well-defined. First, suppose $y_1, y_2 \in M$ both satisfy

$$\beta(y_1) = \beta(y_2) = x.$$

Then $y_1 - y_2 \in \ker\beta = \operatorname{im}\alpha$, so there exists $m' \in M'$ such that

$$y_1 - y_2 = \alpha(m').$$

If $\mu(z_i) = \varphi(y_i)$ for $i = 1, 2$, then

$$\mu(z_1 - z_2) = \varphi(y_1) - \varphi(y_2) = \varphi(\alpha(m')) = \mu(\varphi'(m')),$$

where we used commutativity. Since μ is injective, it follows that

$$z_1 - z_2 = \varphi'(m') \in \operatorname{im} \varphi'.$$

Thus $\overline{z_1} = \overline{z_2}$ in $\operatorname{coker} \varphi'$, so $\delta(x)$ is independent of the choice of y.

Next, if $z, z' \in N'$ both satisfy

$$\mu(z) = \mu(z') = \varphi(y),$$

then $\mu(z - z') = 0$, and since μ is injective, $z = z'$. Hence $\delta(x)$ is also independent of the choice of z.

It is straightforward to verify that δ is a homomorphism.

Now we prove exactness.

Exactness at $\ker \varphi'$. Since α induces α', we have

$$\alpha'(\ker \varphi') \subseteq \ker \varphi.$$

Because α is injective, so is α'. Hence

$$\ker \alpha' = 0.$$

Therefore the sequence is exact at $\ker \varphi'$.

Exactness at $\ker \varphi$. We show that

$$\operatorname{im} \alpha' = \ker \beta'.$$

Take $x' \in \ker \varphi'$. Then

$$\beta'(\alpha'(x')) = \beta(\alpha(x')) = 0,$$

so $\operatorname{im} \alpha' \subseteq \ker \beta'$.

Conversely, let $y \in \ker \varphi$ satisfy $\beta'(y) = 0$. Then $\beta(y) = 0$, hence by exactness of the top row there exists $x' \in M'$ such that

$$\alpha(x') = y.$$

Since $\varphi(y) = 0$, we get

$$0 = \varphi(\alpha(x')) = \mu(\varphi'(x')).$$

As μ is injective, $\varphi'(x') = 0$, so $x' \in \ker \varphi'$. Thus

$$y = \alpha'(x'),$$

proving $\ker \beta' \subseteq \operatorname{im} \alpha'$.

Exactness at $\ker \varphi''$. We show that

$$\operatorname{im} \beta' = \ker \delta.$$

First let $y \in \ker \varphi$. Then $\beta(y) \in \ker \varphi''$, since

$$\varphi''(\beta(y)) = \nu(\varphi(y)) = 0.$$

Thus $\beta'(y) \in \ker \varphi''$. To compute $\delta(\beta'(y))$, choose the lift y itself. Then $\varphi(y) = 0$, so we may take $z = 0$, and hence

$$\delta(\beta'(y)) = \overline{0} = 0.$$

Therefore $\operatorname{im} \beta' \subseteq \ker \delta$.

Conversely, let $x \in \ker \varphi''$ and suppose $\delta(x) = 0$. Choose $y \in M$ with

$$\beta(y) = x,$$

and choose $z \in N'$ such that

$$\mu(z) = \varphi(y).$$

Since $\delta(x) = \overline{z} = 0$ in $\operatorname{coker} \varphi'$, there exists $x' \in M'$ such that

$$z = \varphi'(x').$$

Hence

$$\varphi(y - \alpha(x')) = \varphi(y) - \varphi(\alpha(x')) = \mu(z) - \mu(\varphi'(x')) = 0.$$

So $y - \alpha(x') \in \ker \varphi$, and

$$\beta'(y - \alpha(x')) = \beta(y - \alpha(x')) = \beta(y) = x,$$

because $\beta\alpha = 0$. Thus $x \in \operatorname{im} \beta'$, proving

$$\ker \delta \subseteq \operatorname{im} \beta'.$$

Exactness at $\operatorname{coker} \varphi'$. We show that

$$\operatorname{im} \delta = \ker \mu'.$$

Let $x \in \ker \varphi''$, and write

$$\delta(x) = \overline{z} \in \operatorname{coker} \varphi'$$

as above, with $\mu(z) = \varphi(y)$. Then

$$\mu'(\delta(x)) = \overline{\mu(z)} = \overline{\varphi(y)} = 0$$

in $\operatorname{coker}\varphi$. Hence $\operatorname{im}\delta \subseteq \ker \mu'$.

Conversely, let $\overline{z} \in \operatorname{coker}\varphi'$ satisfy

$$\mu'(\overline{z}) = 0.$$

This means that $\mu(z) \in \operatorname{im}\varphi$, so there exists $y \in M$ such that

$$\varphi(y) = \mu(z).$$

Applying ν, we get

$$\varphi''(\beta(y)) = \nu(\varphi(y)) = \nu(\mu(z)) = 0.$$

Thus $\beta(y) \in \ker \varphi''$. By construction of δ,

$$\delta(\beta(y)) = \overline{z}.$$

So $\overline{z} \in \operatorname{im}\delta$, proving

$$\ker \mu' \subseteq \operatorname{im}\delta.$$

Exactness at $\operatorname{coker}\varphi$. We show that

$$\operatorname{im}\mu' = \ker \nu'.$$

For $\overline{z} \in \operatorname{coker}\varphi'$, we have

$$\nu'(\mu'(\overline{z})) = \overline{\nu(\mu(z))} = \overline{0} = 0,$$

so $\operatorname{im}\mu' \subseteq \ker \nu'$.

Conversely, let $\overline{n} \in \operatorname{coker}\varphi$ satisfy

$$\nu'(\overline{n}) = 0.$$

Then $\overline{\nu(n)} = 0$ in $\operatorname{coker}\varphi''$, so there exists $m'' \in M''$ such that

$$\varphi''(m'') = \nu(n).$$

Since β is surjective, choose $m \in M$ such that

$$\beta(m) = m''.$$

Then

$$\nu(n - \varphi(m)) = \nu(n) - \varphi''(\beta(m)) = \nu(n) - \varphi''(m'') = 0.$$

Hence

$$n - \varphi(m) \in \ker \nu = \operatorname{im}\mu,$$

so there exists $z \in N'$ such that

$$\mu(z) = n - \varphi(m).$$

Therefore in $\operatorname{coker}\varphi$,

$$\overline{n} = \overline{\mu(z)} = \mu'(\overline{z}),$$

showing that $\overline{n} \in \operatorname{im}\mu'$. Thus

$$\ker\nu' \subseteq \operatorname{im}\mu'.$$

Exactness at $\operatorname{coker}\varphi''$ **and termination.** Since $\nu : N \to N''$ is surjective, the induced map

$$\nu' : \operatorname{coker}\varphi \to \operatorname{coker}\varphi''$$

is surjective as well: for any $\overline{n''} \in \operatorname{coker}\varphi''$, choose $n \in N$ such that

$$\nu(n) = n'',$$

then

$$\nu'(\overline{n}) = \overline{n''}.$$

Hence

$$\operatorname{im}\nu' = \operatorname{coker}\varphi'',$$

which is equivalent to exactness at $\operatorname{coker}\varphi''$ and the final 0.

Collecting all the above, we obtain the exact sequence

$$0 \longrightarrow \ker\varphi' \xrightarrow{\alpha'} \ker\varphi \xrightarrow{\beta'} \ker\varphi'' \xrightarrow{\delta} \operatorname{coker}\varphi' \xrightarrow{\mu'} \operatorname{coker}\varphi \xrightarrow{\nu'} \operatorname{coker}\varphi'' \longrightarrow 0.$$

$\square$